NOT MEASUREMENT SENSITIVE

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14 February 1992
SUPERSEDING
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8 March 1985

MILITARY SPECIFICATION

TRUCKS, FLATBED: AND TRUCKS, MAINTENANCE: WITH TELESCOPIC BOOM MOUNTED AERIAL PLATFORM, 8.5 TO 18.3 METERS (28 TO 60 FOOT), MODIFIED COMMERCIAL

This specification is approved for use by the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers diesel and gasoline engine driven, four-wheel, two-rear-wheel drive; four-wheel, four-wheel drive; and six-wheel, four-rear-wheel drive; 6350 to 20 200 kilograms (kg) (14,000 to 44,500 pounds) gross vehicle weight (GVW), commercial trucks equipped with a flatbed or maintenance body and a 8.5 to 18.3 meters (m) (28 to 60 foot) telescopic boom aerial platform. Vehicles procured under this specification are commercial items which are be warranted by the manufacturer as specified in acquisition documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC-2320

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1.2 Classification. The vehicle shall be one of the following body types, as specified (see 6.2):

Type I - Flatbed.
Type II - Maintenance.

1.2.1 Size. The vehicle shall be one of the sizes, classes and models shown in table I. Size of vehicle shall be determined by the maximum height the aerial basket or platform can attain, measured from ground level to the basket or aerial platform floor. Sizes shall be the minimum sizes specified in table I, as specified (see 6.2). Vehicles furnished in a size actually larger than that specified by the procuring activity shall be furnished in the class corresponding to that larger size as indicated in table I.

TABLE I. Vehicle classification.

Size	Class	Model	Ground to aerial basket floor height meters (feet)
28	A	4x2 or 4x4	8.5 (28)
32	B	4x2 or 4x4	9.8 (32)
35	C	4x2 or 4x4	10.7 (35)
45	D	4x2 or 4x4	13.7 (45)
50	E	4x2 or 4x4	15.2 (50)
60	F	6x4	18.3 (60)

1.2.2 Class. The vehicle shall be one of the classes in table II, as specified (see 6.2).

TABLE II. GVW ratings.

Class	GVW rating kilograms (pounds)
A	6350 (14,000)
B	7250 (16,000)
C	8600 (19,000)
D	10 900 (24,000)
E	14 500 (32,000)
F	20 200 (44,500)

1.2.3 Model. The vehicle shall be one of the following models, as specified (see 6.2).

4x2 model - 4 wheels, 2 wheels driving
4x4 model - 4 wheels, 4 wheels driving
6x4 model - 6 wheels, 4 wheels driving.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

	W-B-131	- Battery, Storage: Vehicular, Ignition, Lighting, and Starting.
*	VV-F-800	- Fuel Oil, Diesel

MILITARY

	MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
*	MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
	MIL-B-46176	- Brake Fluid, Silicone, Automotive All Weather, Operational and Preservative.
*	MIL-T-83133	- Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.
	MIL-H-83282	- Hydraulic Fluid, Fire-Resistant, Synthetic Hydrocarbon Base, Aircraft, NATO Code Number H-537.

STANDARDS

FEDERAL

	FED-STD-297	- Rustproofing of Commercial (Nontactical) Vehicles.
	FED-STD-595	- Colors.

MILITARY

	MIL-STD-1223	- Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking, and Data Plate Standards.
	MS 51118	- Pintle Assembly, Towing-40,000 Lbs. Capacity, Manual Release.
	MS 51317	- Light, Warning, Vehicular, Rotating, D.C.
	MS 51335	- Pintle Assembly, Towing - 18,000 Lbs. Capacity, Manual Release.

* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, Military Specifications and Standards, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

* DEPARTMENT OF DEFENSE (DoD)
Department of Defense Index of Specifications and Standards (DODISS).

(Copies of the DODISS are available on a yearly subscription basis either from the Government Printing Office for hard copy, or microfiche copies are available from the Director, Navy Publication and Printing Service Office, 700 Robbins Avenue, Philadelphia, PA 19111-5093.)

DEPARTMENT OF TRANSPORTATION (DoT)
Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

(Application for copies of DoT publications should reference the code of Federal Regulations, 49 CFR, and the Federal Register, and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)
Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

(Application for copies of EPA publications should reference the Code of Federal Regulations, 40 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

* OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)
Occupational Safety and Health Standards.
1926.556 - Aerial Lifts.

(Application for copies of OSHA publications should reference the Code of Federal Regulations, Title 29, and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

* 2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI A92.2	- American National Standard for Vehicle-Mounted Elevating and Rotating Aerial Devices.

(Application for copies of ANSI publications should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A120	- Pipe, Steel, Black and Hot-Dipped, Zinc-Coated (Galvanized) Welded and Seamless for Ordinary Uses (DOD adopted).
ASTM B117	- Standard Method of Salt Spray (Fog) Testing (DOD adopted).
* ASTM D 4985	- Standard Specification for Low Silicate Ethylene Glycol Base Engine Coolant for Heavy Duty Engines Requiring an Initial Charge of Supplemental Coolant Additive.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

* THE EUROPEAN TYRE AND RIM TECHNICAL ORGANISATION (ETRTO)
Standards Manual

(Application for copies of the ETRTO publications should be addressed to the European Tyre and Rim Technical Organisation, 32, Avenue Brugmann, 1060 Brussels, Belgium.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA No. 70	- National Electric Code.

(Application for copies of NFPA publications should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

SAE, INC.
SAE Standards and Recommended Practices.

J318	- Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers - Trucks, Truck-Tractors, and Trailers (DoD adopted).
J350	- Spark Arrester Test Procedure for Medium Sized Engines (DoD adopted).
J537	- Storage Batteries (DoD adopted).
J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (30-1000 MHz).

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J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
J588	- Turn Signal Lamps.
J589	- Turn Signal Switch.
J682	- Rear Wheel Splash and Stone Throw Protection (DoD adopted).
J683	- Tire Chain Clearance, Trucks, Buses, and Combinations of Vehicles (DoD adopted).
J688	- Truck Ability Prediction Procedure (DoD adopted).
J704	- Openings for Six- and Eight-Bolt Truck Transmission Mounted Power Take-Offs (DoD adopted).
J706	- Rating of Winches.
J844	- Nonmetallic Air Brake System Tubing (DoD adopted).
J1067	- Seven-Conductor Jacketed Cable for Truck-Trailer Connections.
J1292	- Automobile, Truck, Truck-Tractor, Trailer, and Motor Coach Wiring.
J1349	- Engine Power Test Code - Spark Ignition and Diesel.

(Application for copies of SAE publications should be addressed to the SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Year Book.

* (Application for copies of the Tire and Rim Association publications should be addressed to the Tire and Rim Association, Inc., 175 Montrose West Ave., Copley, OH 44321.)

* (Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.15, the vehicle, components, assemblies, and accessories to be delivered under the contract as further identified and qualified in the

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product experience clause of the solicitation shall be the standard and optional items sold commercially to the public which meet or exceed the requirements of this specification. Except as specified in 3.1.1 through 3.1.1.15, no removal, substitution or alteration of the chassis manufacturer's standard or optional chassis model components shall be made. All chassis items shall be as represented and rated in the chassis manufacturer's technical data book. Special bodies, mounted telescopic boom, aerial platform and equipment shall be as represented in the body and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material identical to that furnished to and utilized by the authorized company representatives for selection of vehicle models and components and shall be available to the engineering offices of the procuring activity, prior to delivery of the items. The chassis model furnished shall be not older than the chassis manufacturer's current model on the date of invitation for bids.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

* 3.1.1.1 Treatment and painting. The vehicle shall be treated and painted in accordance with MIL-STD-1223. The exterior color shall be gloss yellow, matching color chip No. 13538 of FED-STD-595. Extruded, molded or spiral wound booms and baskets shall not be painted.

* 3.1.1.2 Marking and data plates. As specified by the procuring activity for the appropriate military service (see 6.2), identification marking and data plates shall be in accordance with MIL-STD-1223. When specified (see 6.2), concealed markings shall be furnished.

* 3.1.1.2.1 Warning plates. Warning plates shall be furnished and mounted in a conspicuous location near the lower and upper controls and in the cab. Plates shall conform to composition A (class 1 or 2) or composition C of MIL-P-514. At least the following information shall appear on each of the three plates:

WARNING

(a) AERIAL PLATFORM AND OUTRIGGERS MUST BE IN STOWED POSITION WHILE VEHICLE IS BEING MOVED.
(b) RATED PLATFORM LOAD, HYDRAULIC OUTRIGGERS DOWN, NOT IN EXCESS OF _____ POUNDS.
(c) INSPECT PLATFORM LEVELING SYSTEM.
(d) MAXIMUM VOLTAGE BOOM AND PLATFORM WILL WITHSTAND _____ VOLTS AC.
(e) OVERALL TRAVEL HEIGHT WHEN BOOM IS IN STOWED POSITION _____.
(f) VEHICLE SHALL NOT BE OPERATED IN WINDS IN EXCESS OF 48 KILOMETERS PER HOUR (KM/H) (30 MPH).
(g) BASKET LINER(S) MUST BE INSTALLED WHEN OPERATING ON OR NEAR ELECTRICAL LINES.

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(h) DO NOT PERMIT DIRECT CONTACT OF BASKET TO GROUND WHEN OPERATING ON OR NEAR ELECTRICAL DISTRIBUTION LINES.
(i) MAXIMUM LONGITUDINAL INCLINE _____ DEGREES.
(j) MAXIMUM TRANSVERSE SLOPE _____ DEGREES.
(k) BOOM AND BASKET(S) MUST BE KEPT CLEAN AND FREE OF DAMPNESS.
(l) DO NOT DRILL HOLES IN BOOM OR BASKET TO DRAIN WATER, DUMP OUT WATER.
* (m) USE ONLY RECOMMENDED TYPE OF HYDRAULIC FLUID.
* (n) WIRE BRAID HYDRAULIC HOSES MUST NOT BE USED ON OR NEAR ELECTRICAL LINES.

* 3.1.1.2.2 Instruction plates. An instruction plate showing a schematic of all valves, valve controls and equipment, and complete warning and instructions for operating the aerial equipment shall be fastened on the inside of a curbside vertical cabinet door. Each control lever shall be marked with instruction plate depicting the function of the control. Individual markings shall be adjacent to each control. Plates shall conform to composition A (class 1 or 2) or composition C of MIL-P-514.

3.1.1.2.3 Warning signs. Warning signs or labels, as recommended by the Manufacturers of Aerial Devices and Digger Derrick Council, shall be furnished and attached.

3.1.1.3 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with FED-STD-297. When specified (see 6.2), tropical rustproofing in accordance with FED-STD-297 shall be furnished.

3.1.1.4 Drain plugs. All drain plugs installed in manual transmission, transfer case, and rear axle for draining lubricants shall be of the permanent magnet type.

* 3.1.1.5 Towing devices. Not less than two hooks, loops, pins or a single, center mounted towing eye, mounted to the frame or to rigid members which are attached to the frame, for towing the vehicle, shall be furnished on the front of the vehicle. All towing devices shall be frame rail mounted or reinforced back to each frame rail.

* 3.1.1.6 Trailer towing package. A trailer towing package consisting of a pintle, safety chain attachment devices, a lighting receptacle and associated reinforcements and wiring shall be installed on the rear of the vehicle. The pintle shall be of the rotating type conforming to MS 51335-1 for 4x2 and 4x4 model vehicles and MS 51118 for 6x4 model vehicles. The pintle shall be installed on the chassis frame with reinforcements to transfer a vertical tongue load of not less than 815 kg (1,800 pounds) and a horizontal drawbar load of not less than 8150 kg (18,000 pounds) for 4x2 and 4x4 model vehicles and a vertical tongue load of not less than 1815 kg (4,000 pounds) and a horizontal drawbar load of not less than 178 kilonewtons (kN) (40,000 pounds) for 6x4 model vehicles directly to the chassis rails. The rearmost portion of the pintle shall be forward but not more than 100 mm

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(four inches) forward of the rearmost part of the vehicle. Two trailer safety chain attachment devices, one adjacent to each side of the pintle, shall be provided. Each attachment device shall provide an ultimate strength at least equal to the GVW of the truck furnished. The safety chain attachment devices shall be capable of accommodating a standard grab hook [56 millimeters (mm) (2-3/16 inch) wide, 18.7 mm (47/64 inch) thick and 12.3 mm (31/64 inch) throat width)] for a 9.5 mm (3/8-inch) chain for 4x2 and 4x4 model vehicles and [116 mm (4-9/16 inches) wide, 30 mm (1-3/16 inches) thick, 19.8 mm (25/32 inch) throat width)] for a 16 mm (5/8-inch) chain for 6x4 model vehicles. The lighting receptacle, conforming to SAE J560 with its conductors connected and color coded as specified therein, shall be mounted in a readily accessible location near the pintle.

3.1.1.7 Trailer lighting cable. When specified (see 6.2), a coiled electrical trailer lighting cable conforming to SAE J1067 shall be furnished. The coiled cable shall be not less than 2800 mm (110 inches) long when fully extended. Both ends of the cable shall be equipped with a seven-circuit round plug conforming to SAE J560. The plugs shall be equipped with a grip for withdrawing from the connector sockets. The cable shall be packaged and stowed in the vehicle tool compartment.

3.1.1.8 Electric trailer brake controller. When specified (see 6.2), the vehicle shall be provided with an electric trailer brake controller. The controller shall be installed in the truck cab and wired through the SAE J560 lighting receptacle. The controller shall operate on the 12-volt electrical system of the vehicle and shall include load control, hand lever and accessories.

3.1.1.9 Trailer air brake system. When specified (see 6.2), a trailer air brake system shall be furnished. The trailer air brake system shall include full air brakes as specified in 3.4.11.2 and the trailer brake control system specified in 3.4.11.3.

* 3.1.1.10 Spark arrester. As specified herein (see 3.4.4), the vehicle shall be furnished with an exhaust system spark arrester, unless the engine is turbocharged.

* 3.1.1.11 Rear wheel splash and stone throw protection. Rear wheel splash and stone throw protection shall be in accordance with provisions of SAE J682. A metal strip not less than 3.2 mm (0.125 inch) thick and not less than 25 mm (one inch) wide, extending the entire width of the mud flap, shall be installed to prevent the bolt heads or bolt nuts from damaging the mud flap. As an alternate method of attaching the mud flaps, tabs or clips with a minimum surface contract dimensions of 25 mm (1 inch) high by 32 mm (1.25 inch wide) by 2.4 mm (0.094 inch) thick shall be furnished at each bolt.

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* 3.1.1.12 Brake lights. At least one pair of brake lights shall override the four-way emergency flasher or the two systems shall be independent of each other. Modifications to the manufacturer's standard product to accommodate this requirement shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard. If additional lights are added to the vehicle, the lights shall be selected from the chassis manufacturer's standard matching hardware.

* 3.1.1.13 4x4 conversion. The chassis manufacturer's standard 4x2 truck chassis may be modified to provide all wheel drive conforming to the requirements specified herein if:

(a) The conversion axle manufacturer's engineering department specifically approves and certifies that all such modifications meet the design requirements and standards of the conversion axle manufacturer. Certification shall be based on both design analysis and proving ground test reports, which shall be made available to the engineering and quality assurance offices of the procuring activity.
(b) The chassis manufacturer's front axle before conversion has the same load rating as the conversion axle to be installed.
(c) Components used in the all wheel drive conversion are of current production.
(d) Components used in the all wheel drive conversion are approved for the conversion application by the component manufacturers.
(e) The converted vehicle is certified to conform to Federal Motor Vehicle Safety Standard No. 121, by the intermediate or final manufacturer.
* (f) Replacement headlights, if required to be added, shall meet the height requirement of not less than 560 mm (22 inches) and not more than 1370 mm (54 inches), measured above the road surface, in conformance with Federal Motor Vehicle Safety Standard No. 108. Replacement headlights shall be equivalent in mounting, protection, and range and precision of adjustment to the chassis manufacturer's original standard headlights.
(g) Unused headlight cavities are covered in a neat workmanlike manner, treated and painted to match the chassis cab color with treatment and painting equivalent to the chassis cab manufacturer's process for the remainder of the chassis cab. Cavities and their covers shall be rustproofed in accordance with 3.1.1.3.
(h) Complete installation drawings are available to the procuring activity.

3.1.1.14 Silicone brake fluid. When specified for class A, B and C vehicles (see 6.2), and if available as the manufacturer's standard or optional brake fluid, brake fluid conforming to MIL-B-46176 shall be provided

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in the hydraulic brake system. A tag shall be placed near the master cylinder stating "CAUTION: USE SILICONE BRAKE FLUID ONLY, MIL-B-46176."

* 3.1.1.15 Brushguard. Model 4x4 vehicles shall be equipped with a radiator and headlamp brushguard. When the headlamps are recessed into and protected by the front bumper, a headlamp brushguard is not required.

3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

3.2.2 Air pollution control. The vehicle shall comply with EPA regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

* 3.2.3 Sound level. The cab interior sound level shall not exceed 84 db(A) when measured in accordance with Federal Motor Carrier Safety Regulation 393.94. The vehicle exterior drive-by sound level shall conform to EPA Noise Emission Standards for Transportation Equipment, Medium and Heavy Trucks.

* 3.2.4 Safety. The vehicle and rotating aerial devices shall comply with ANSI A92.2 and all OSHA requirements, including OSHA 1926.556.

3.2.5 Curb weight. The curb weight shall include the weight of the chassis and cab, with all attachments, accessories, and equipment; body; basket or aerial platform assembly in stowed position; and a full complement of fuel, lubricants and coolant.

3.2.6 Gross vehicle weight. The gross vehicle weight (GVW) shall consist of curb weight, operator weight (computed at 80 kg (175 pounds)) and a payload uniformly distributed throughout the length of vehicle body, to provide not less than the specified GVW. A vehicle with a crew (four-door) cab shall have the GVW computed with 240 kg (525 pounds) of payload in the rear seat.

3.2.7 Weight distribution. The axle capacity and capacity of suspension and tires per axle shall be not less than as specified in table III.

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TABLE III. Suspension, axle and tire capacities.

Class	Size	Front axle kg (pounds)	Rear axle(s) kg (pounds)	GVW kg (pounds)
A	28	2270 (5,000)	4540 (10,000)	6350 (14,000)
B	32	2720 (6,000)	5450 (12,000)	7250 (16,000)
C	35	3180 (7,000)	6350 (14,000)	8600 (19,000)
D	45	4085 (9,000)	8360 (18,500)	10 900 (24,000)
E	50	5450 (12,000)	9310 (20,500)	14 500 (32,000)
F	60	5450 (12,000)	15 400 (34,000)	20 200 (44,500)

3.2.8 Ratings. Vehicle ratings shall be the manufacturer's published rating. Component and vehicular ratings shall not be raised to meet the requirements of this specification. Minimum GVW ratings shall be not less than specified in table II for the specified class of vehicle.

3.2.9 Cab-to-axle. The cab-to-axle or cab-to-trunnion dimension shall be not less than 1520 mm (60 inches) for class A, not less than 2108 mm (83 inches) for classes B, C, and D and not less than 3050 mm (120 inches) for classes E and F. The cab-to-axle or cab-to-trunnion dimension shall be measured from back of cab or crew cab, as applicable, to the center of the axle or to the center of the trunnion.

3.2.10 Overall width. The overall width of the vehicle, exclusive of mirrors, lights, reflectors and tires, shall be not more than 2440 mm (96 inches). The width over the tires shall be not more than 2540 mm (100 inches).

* 3.2.11 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

* 3.2.12 Prohibited materials. Asbestos materials shall not be used in any form in any part of the vehicle. No item, part or assembly shall contain radioactive materials in which the specific activity is greater than 0.002 microcurie per gram or activity per item equals or exceeds 0.01 microcuries.

3.3 Performance.

3.3.1 Speeds and gradeability. High and low speed requirements shall be met with vehicle loaded to specified GVW.

3.3.1.1 High speed gradeability. The vehicle shall ascend continuous grades specified in table IV at 80 km/h (50 miles per hour (mph)). Gradeability requirements shall be met with the main transmission in direct

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drive, and when a multi-speed axle or auxiliary transmission is specified, with the axle or auxiliary transmission in high speed range. In addition, 4x4 model performance requirements shall be met with the front wheel drive disengaged, except that when equipped with interaxle compensating devices, the vehicle shall meet the performance requirements with front wheel drive engaged. Gradeability shall be verified with calculations in accordance with SAE J688 (see 6.3).

* TABLE IV. Gradeability.

Vehicle class	Percent of grade
A	3.0
B	3.0
C	3.0
D	2.5
E	1.5
F	1.1

3.3.1.2 Low speed. Low speed for vehicles with a manual transmission shall be calculated with the engine operating at not less than 35 percent of recommended governed speed, and shall provide a vehicle speed of not more than 6.4 km/h (4 mph) for class A, B and C vehicles and not more than 4.8 km/h (3 mph) for class D, E and F vehicles.

* 3.3.1.3 Maximum geared speed. The maximum geared speed at engine governed speed shall be not less than 84 km/h (52 mph) for model 4x4 vehicles and not less than 93 km/h (58 mph) for model 4x2 and model 6x4 vehicles. Conformance to geared speed specified shall be determined by calculating in accordance with the following formula:

$$\text{Maximum geared speed (km/h)} = \frac{\text{Governed speed (rpm)} \times 1.609}{\text{Total gear reduction} \times \text{tire factor (see 6.3)}}$$

3.3.2 Service brakes. The service brakes shall control and hold the vehicle when loaded to its specified GVW, on a 30 percent grade. The service brakes shall stop the vehicle, loaded to specified GVW, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52.

* 3.3.3 Upsetting stability. With one and one-half times the specified load (see 3.6.5 and 3.6.5.2) placed in or suspended from the basket, with the vehicle otherwise unloaded, and with all the outriggers extended, the telescopic boom shall be capable of placing and holding the loaded basket in any attainable position of the equipment without causing the vehicle to become unstable. The vehicle shall be deemed unstable with one and one-half times the specified load when any one of the vehicle wheels or outriggers lifts off the ground, ANSI A92.2 not withstanding. The stability test shall

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not cause any permanent deformation beyond the elastic limit of any component of the vehicle. The vehicle shall also be deemed unstable if, with twice the specified load (see 3.6.5), the inner tire of either rear wheel lifts off of the ground, ANSI A92.2 not withstanding.

* 3.3.3.1 Stability on slopes. With one and one-third times the specified load (see 3.6.5 and 3.6.5.2), placed in or suspended from the basket, with the vehicle otherwise unloaded, with the outriggers extended, and with the vehicle not being leveled (ANSI A92.2 not withstanding), on a 5 degree slope, facing upward and then downward, and then at right angles to the 5 degree slope, the telescopic boom shall be capable of placing and holding the loaded basket in any attainable position of the equipment without causing the vehicle to become unstable. The vehicle shall be deemed unstable with one and one-third times the specified load when any one of the vehicle's tires or outriggers lifts off of the ground, ANSI A92.2 not withstanding.

3.3.4 Wind resistance. With the specified load (see 3.6.5 and 3.6.5.2) in the basket or aerial platform, with the vehicle otherwise unloaded, and in any extended position, the vehicle shall be capable of withstanding a wind load of 335 kilograms per square meter (kg/m^2) (7 pounds per square foot) without upsetting, overturning, or permanent deformation of any component. The maximum area exposure for wind load shall be calculated with the body, boom, and basket or aerial platform in the most critical position that would induce tipping. The resisting moment shall be greater than the wind load when the load is applied on the surface of the body, boom, and basket or aerial platform. The contractor shall perform calculations or shall have available a certified test report for the vehicle or a vehicle of the same class and size.

3.3.5 Height. The boom shall be capable of raising the basket or aerial platform with rated load (see 3.6.5 and 3.6.5.2), measured from ground level to the basket or aerial platform floor, to the minimum height for the size of the vehicle specified in table I.

3.3.6 Horizontal reach. The horizontal reach shall be not less than that specified in table V, measured from the front edge of the basket or aerial platform to the axis of the turntable. The boom shall be capable of lowering the loaded basket or aerial platform to the ground. The boom shall be capable of being elevated from not less than 10 degrees below horizontal to not less than 70 degrees above horizontal with rated payload and operating at normal rated speed.

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TABLE V. Horizontal reach.

Class	Sizes	Horizontal reach meters (feet)
A	28	6.4 (21)
B	32	7.3 (24)
C	35	8.2 (27)
D	45	11.0 (36)
E	50	12.5 (41)
F	60	13.7 (45)

* 3.3.7 Rotation (boom assembly). The boom assembly shall be capable of continuous rotation in either direction in the horizontal plane. The turntable motor shall rotate the loaded platform through an arc of 360 degrees in not less than 60 seconds and not more than 120 seconds. The turntable rotating device shall be capable of remaining in fixed position when the hydraulic motor is de-energized. The turntable shall be capable of smoothly rotating the aerial assembly up and down a 5 percent grade when loaded to rated working capacity, and producing the maximum attainable tipping moment.

3.3.8 Position stability. When the vehicle is on a 5 percent grade, without being leveled, the elevating and rotating mechanism shall hold the loaded basket or aerial platform (see 3.6.5 and 3.6.5.2) in any extended position within 25 mm/h (1 inch per hour) horizontal creep and 50 mm/h (2 inches per hour) vertical settling. The creep and settling shall be measured at the front of the basket or aerial platform, with hydraulic fluid stabilized at ambient air temperature.

3.4 Chassis components.

3.4.1 Engine. The engine furnished for the specified vehicle class shall be the chassis manufacturer's standard or optional engine for the commercial model truck which meets or exceeds the requirements of this specification.

* 3.4.1.1 Diesel engine. Unless otherwise specified (see 3.4.1.2), the vehicle shall be equipped with a liquid cooled, compression ignition, two-stroke or four-stroke cycle diesel engine. Engine net power used in performance prediction calculations shall be determined in accordance with or SAE J1349. The engine shall demonstrate the performance characteristics specified herein when using diesel fuel conforming to VV-F-800. In addition, when specified (see 6.2), the engine shall operate satisfactorily on grade JP-4 and grade JP-5 fuel conforming to MIL-T-5624 under emergency, short duration conditions and on grade JP-8 fuel conforming to MIL-T-83133 under normal conditions. A power loss when operating on JP-4, JP-5 or JP-8 is acceptable.

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* 3.4.1.2 Gasoline engine. When specified (see 6.2), the vehicle shall be equipped with a liquid cooled, internal combustion, four-stroke cycle gasoline engine. The engine furnished shall produce the required vehicle performance when operated on unleaded fuel with a research octane rating of 91, at engine speed not more than the manufacturer's recommended operating speed. The engine shall be capable of warranted operation on unleaded fuel when used in accordance with the operator's manual. Net power figures used in performance calculations shall be determined in accordance with or SAE J1349. Unless otherwise specified or required by foreign laws, vehicles for overseas use shall be capable of accepting and operating on leaded gasoline. Class D, E and F vehicles shall be provided with a fan clutch.

3.4.1.3 Oil filter. A full flow type oil filter shall be furnished.

3.4.1.4 Governor. For class C, D, E and F vehicles, an engine governor shall be furnished and set and sealed to limit the engine to the engine manufacturer's maximum recommended operating speed.

* 3.4.1.5 Cooling system. The cooling system shall maintain the engine coolant at a temperature below the boiling point with the vehicle loaded to GVW and operated at an altitude of 3050 meters (m) (10,000 feet) above sea level or in an ambient air temperature of not less than 52° Celsius (°C) (125 degrees Fahrenheit (°F)). The coolant system shall include a surge tank or a coolant recovery reservoir of not less than 1.89 liter (L) (two-quart) capacity. For cooling system servicing, see 3.4.25.1.

* 3.4.1.6 Fan clutch. A fan clutch shall be provided. The fan clutch shall reduce the fan speed automatically when the fan is not required for engine cooling. The fan clutch shall be asbestos free.

* 3.4.1.7 Coolant temperature control. Thermostatic control of engine coolant temperature shall be provided. On diesel engine driven vehicles, the control shall include complete thermostatic control of all flow through the radiator.

3.4.1.8 Silicone rubber hoses. When specified (see 6.2), silicone rubber radiator and heater hoses shall be furnished.

* 3.4.1.9 Power plant heaters and fuel warmer. When specified (see 6.2), a coolant heater; an engine oil heater; and fuel warmers (diesel engine driven vehicles only) shall be provided. Heaters shall operate on 110-volt alternating current (ac), and shall be wired through a junction block, including a fuse or circuit breaker, to a single, three-pronged (male), weatherproof, slave receptacle for receiving external power and grounding vehicle. The receptacle shall be located on the streetside of the vehicle, as near the cab door as possible. A three-wire connecting cable, not less than 7600 mm (25 feet) long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. The connecting cable

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shall include a matching female connector at the vehicle end and a standard weatherproof, three-pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation 393.77(c)(7). Electrical insulation of the connecting cable shall withstand normal operating stresses in low ambient air temperatures (down to -51°C (-60°F)) without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation, and without loose excess wire. Provisions for stowage of the cable shall be provided in the vehicle cab. Heaters and fuel warmers shall be furnished as follows:

(a) A coolant heater, 1500-watt (W) minimum rating, shall be installed in the engine block or lower coolant inlet hose. An engine thermostat with an operating range of 77°C (170°F) to 90°C (195°F) shall be installed.

* (b) An oil pan heater of the permanent external surface mount, permanent in-pan mount, or immersion type that meets the following requirements, shall be installed:

* (1) Not less than 16 W/L (15 watts per quart) heating capacity

* (2) Not more than 2.8 watts per square centimeter (W/cm^2) (18 watts per square inch) heating capacity

(3) Thermal balance design or thermostat control providing for uninterrupted operation

(4) Provision for mounting below minimum service oil level.

* (c) An in-line fuel warmer or preheater unit shall be provided on diesel engine driven vehicles to prevent clogging of fuel filters due to wax crystallization in the fuel. The unit shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of -40°C (-40°F) to an outlet temperature of -13°C (+9°F), with a fuel flow rate not less than the maximum fuel demand of the engine fuel system. A coolant shutoff valve shall be provided for the coolant inlet side of the fuel warmer unit. The unit shall not cause heating of the fuel above 27°C (80°F) under any possible condition.

* (d) An in-tank fuel warmer or preheater unit shall be provided on diesel engine driven vehicles. The unit shall use engine coolant to transfer heat to the fuel in one fuel tank. The warmer shall not cause heating of any fuel above 27°C (80°F) under any possible condition, shall not disable or cause elimination of the fuel gage sending unit and shall not violate 3.2.1 or any Federal Motor Carrier Safety Regulation. A coolant shut off valve shall be included. The units required by 3.4.1.9(c) and (d) may be combined.

(e) When an auxiliary engine (see 3.6.9) is provided, a 300-watt immersion type heater with 77°C (170°F) to 90°C (195°F) thermostat shall be installed in the oil pan through any convenient opening.

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* 3.4.1.10 Vernier throttle control. When specified (see 6.2), a manual vernier controlled throttle with quick release shall be furnished, except on electronically controlled engines.

3.4.2 Electrical system. The electrical system shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

* 3.4.2.1 Starting system. For diesel engine driven vehicles, a 12- or 24-volt direct current (dc) starting system, with a 12-volt dc lighting system shall be furnished. Engine starting equipment shall include an ether starting system, glow plug or grid heater. If an ether system is furnished in lieu of a glow plug or grid heater, it shall be of the measured shot type. The measured shot type ether system shall be key operated or manually operated from the driver's compartment and shall be inoperative with the engine warm. Complete provisions for a replaceable ether reservoir of not less than 355 milliliters (mL) (12 fluid ounces) shall be furnished. A reservoir need not be furnished. When specified (see 6.2), the starting motor shall be equipped with a thermostat controlled, automatic resetting circuit breaker that will protect the motor from overcrank heat damage. The circuit breaker shall not limit cranking ability for a startable engine.

* 3.4.2.2 Ignition system. For gasoline engine driven vehicles, a 12-volt dc ignition system shall be furnished.

* 3.4.2.3 Alternator. Unless otherwise specified (see 6.2), a minimum 85-ampere alternator shall be furnished. The alternator output with the engine at engine idle speed shall be not less than 45 amperes.

3.4.2.4 Lighting. All vehicle lights, reflectors and wiring shall conform to Federal Motor Carrier Safety Regulations 393.12, 393.19, 393.20 and 393.22 through 393.26(d). Lights and reflectors shall not be mounted on vertical surface of rub rails (unless recessed and fully protected) or mounted on vehicle bumpers.

* 3.4.2.5 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589, class A, and shall be mounted on the steering column. Turn signal units shall be installed in accordance with SAE J588. Turn signal operating units shall have a visible flash indicator.

3.4.2.6 Batteries. Each battery shall be of 12-volt potential. The total reserve capacity rating and the total cold cranking rating at -18°C (0°F), both measured in accordance with SAE J537, shall be not less than as specified in table VI. Batteries shall be of the maintenance-free type having the maintenance-free characteristics listed in W-B-131.

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* TABLE VI. Batteries.

Engine type	Reserve capacity (minutes)	Cold cranking (amperes)
Diesel, 200 kilowatts (kW) gross (270 gross horsepower) or less	480	1,875
Diesel, over 200 kW gross (270 gross horsepower)	640	2,500
Gasoline	110	450

3.4.2.7 Radio interference suppression. The vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551. Any body equipment emitting radio interference shall be suppressed to the same level as the vehicle chassis.

3.4.3 Fuel system. The fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67.

* 3.4.3.1 Air cleaner. An air cleaner shall be furnished.

* 3.4.3.2 Fuel tank(s). The vehicle shall be equipped with fuel tank(s) of not less than 189 L (50 gallons) total capacity (163 L (43 gallons) for vehicles destined for California). When more than one tank is furnished on diesel engine driven vehicles, means shall be provided to assure an equalized fuel level in both tanks. When more than one tank is furnished on gasoline engine driven vehicles, a selector valve, connecting either tank to the engine fuel intake shall be provided and means shall be provided to monitor the fuel level of either tank from a single fuel gage, or an equalizing pump shall be used to maintain the same fuel level in both tanks.

3.4.4 Exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83. A spark arrester with an efficiency rating of not less than 80 percent when tested in accordance with SAE J350 shall be furnished, except on turbocharged engines. Vertical exhaust mufflers, if furnished and capable of being reached easily by personnel entering or leaving either side of the cab, shall be provided with a heat shield. Vertical exhaust systems shall be provided with a hinged rain cap.

3.4.4.1 Exhaust system safety shield. The exhaust system or hydraulic system shall be shielded so that, in the event of a hydraulic line rupture or hydraulic fluid leak, the hydraulic fluid will not come into contact with any part of the vehicle exhaust system.

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3.4.5 Gear train.

3.4.5.1 Transmission. Unless otherwise specified (see 3.4.5.3), a manually shifted transmission shall be provided. Input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. The transmission shall be provided with a power takeoff opening conforming to SAE J704. The transmission shall provide for maximum ease of shifting in all speeds (see 3.4.5.4).

* 3.4.5.1.1 Speeds. The vehicle shall be provided with the number of forward speeds specified in table VII. Gear ratios in the transmission shall be matched to provide a progressive shifting pattern throughout the complete range, and shall provide the vehicle performance specified in 3.3.1. through 3.3.1.3.

TABLE VII. Vehicle requirements, minimum number of forward speeds.

Vehicle model	Number of forward speeds
4x2	5
4x4	9
6x4	13

* 3.4.5.1.2 Transfer case (4x4 model). Model 4x4 vehicles shall be provided with a transfer case. Unless the transfer case is equipped with devices which compensate for differential torque and speeds between the front and rear axles, the transfer case shall provide for driver selection of either two-wheel or four-wheel drive. When interaxle compensating devices are furnished, they shall provide for positive transfer of power to all the driving axles. When a two-speed transfer case is furnished, the speedometer shall read accurate speed km/h (mph) with the speed selector in high or low range.

* 3.4.5.2 Clutch. The clutch shall be the largest capacity clutch offered for the class and model vehicle and size of engine furnished with clutch torque capacity exceeding maximum delivered engine torque. The clutch lining shall be asbestos free. The clutch on model 6x4 shall have spring dampening and a greaseable bearing.

* 3.4.5.3 Automatic transmission. When specified (see 6.2), the vehicle shall be provided with an automatic transmission. The transmission shall provide continuous drive. The transmission shall include a hydraulic torque converter and not less than four forward gear ratios for class A, B, C, D and E vehicles, and not less than 5 forward gear ratios for class F vehicles. Normal driving range selector position shall provide not less than four gear

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ratios without movement of the selector. The transmission shall be provided with a power takeoff opening. The input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine.

3.4.5.4 Power takeoff. When a power takeoff is furnished, it shall be of a rated capacity to operate all auxiliary equipment. Controls to operate the power takeoff shall be located in the truck cab accessible to the seated driver. A caution plate or decal reading "DO NOT OPERATE VEHICLE AT HIGHWAY SPEEDS WITH POWER TAKEOFF ENGAGED" shall be installed so as to be readily visible to the seated driver.

3.4.6 Driveline components. Driveline components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction.

3.4.7 Frame. The chassis frame shall be the manufacturer's standard for the type of vehicle furnished. Model 4x2 and 4x4 vehicles shall have a frame with reinforcements extending at least from the rear of the front suspension rear hanger bracket to the rear spring front hanger bracket. Model 6x4 vehicles shall have a frame with reinforcements extending at least from the rear of the front suspension rear hanger bracket to the front of the bogie trunnion mounting bracket. Reinforcements shall provide sufficient structural strength in the chassis frame through increased bending moment to at least equal the loads imposed with the vehicle body loaded to provide specified GVW and with the loaded telescopic boom moving through all attainable positions.

3.4.8 Suspension. The vehicle shall be equipped with a suspension system with components having a total rated capacity per axle at least equal to that specified in 3.2.6. The vehicle shall be equipped with hydraulic, double-acting shock absorbers at the front wheels. Suspension system ratings shall be adequate for the loads imposed during all operating conditions of the aerial booms.

* 3.4.9 Axles. Axle ratings shall be at least equal to that specified in 3.2.6. Gear ratio(s) shall provide performance specified in 3.3.1. Except for front drive axles on model 4x4, the wheel bearings and axle spindles shall be oil lubricated. The hubcaps, except for driving axles, shall have a window for visual determination of oil level. Provisions for venting or withstanding internal pressure buildup and for replenishing the oil supply shall be provided.

3.4.9.1 Rear bogie. The rear bogie on 6x4 model vehicles shall be of the four-wheel type, complete with axles, springs, torque rods and all other necessary parts. The bogie shall be provided with means permitting differential action between the two axles, and a manually or automatically controlled lockout assuring equal power to each rear axle. The manual lockout control used shall be located in the truck cab. Axle gear ratios shall provide performance specified in 3.3.1 through 3.3.1.3.

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3.4.9.2 Two-speed axle. When specified for 4x2 and 6x4 model vehicles (see 6.2), a two-speed axle shall be furnished. The axle shall be equipped with electric, vacuum, or air shift and shall be provided with ratios which will permit proper transmission gear splitting. The gear ratios shall provide the performance specified in 3.3.1 through 3.3.1.3.

3.4.9.3 Traction control. When specified (see 6.2), the vehicle shall be furnished with a traction control on the rear axle. For 6x4 model vehicles, traction control shall be on the forward rear axle. The traction control shall actuate automatically to insure that power is transmitted to the wheel(s) having traction when the opposite wheel(s) loses traction. Maximum traction capabilities shall be maintained at all times under each drive wheel for the life of the vehicle.

* 3.4.10 Wheels, rims, tires, and tubes. Unless otherwise specified, the vehicle shall be equipped with single front and dual rear wheels. Rims and tire ratings shall conform to Tire and Rim Association or European Tyre and Rim Technical Organization recommendations for the type and size of tires furnished. Except when tube type tires are specified in the procurement documents, multi-piece rims shall not be furnished on front axles rated at 6350 kg (14,000 pounds) or less and on single rear axles rated at 10 430 kg (23,000 pounds) or less or tandem axles rated at 20 900 kg (46,000 pounds) or less. Tire, rim size, disc wheel size, and ply rating shall be the same for all wheels on each vehicle. When specified (see 6.2), the vehicle shall be equipped with wide base type front and rear tires and wheels in lieu of single front and dual rear wheels. All wide base wheels, including the spare wheel, shall be interchangeable between the front and rear axle(s) without the use of an adapter. When specified (see 6.2), disc type wheels shall be furnished.

* 3.4.10.1 Tires. Tires shall be tubeless type with highway tread for models 4x2 and 6x4 and with nondirectional mud and snow tread for model 4x4 vehicles. Tires for class C, D, E and F vehicles shall be of the steel belted radial type or, when specified (see 6.2), bias ply. Unless otherwise specified, low profile or standard profile tires may be furnished. When specified (see 6.2), low profile tires shall be furnished. Tires shall be of rated capacity at least equal to that specified in 3.2.6. Tires shall conform to Tire and Rim Association or to the European Tyre and Rim Technical Organisation recommendations.

* 3.4.10.2 Inner tubes. When tube type tires are specified in acquisition documents, inner tubes shall be of heavy duty type, and shall be of proper size for tires furnished. Tube type radial tires shall have radial tire inner tubes. Tire flaps shall be provided for tube type tires in accordance with Tire and Rim Association recommendations.

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* 3.4.10.3 Carrier for spare tire assembly. When specified (see 6.2), a carrier for a spare wheel or rim and tire assembly shall be installed in a readily accessible location on the vehicle. Threaded fasteners, when used to secure the spare tire in the carrier, shall be constructed of or plated with corrosion-resistant material. The carrier design shall enable removal or mounting of the spare wheel assembly using only the tools specified in 3.4.16.1. The carrier shall enable the safe removal and installation of the spare tire assembly from and to the vehicle and carrier without personnel positioning themselves or any part of their body under the spare tire assembly.

* 3.4.10.4 Spare wheel or rim. When specified (see 6.2), a spare wheel or rim identical to those furnished on the front axle of the vehicle shall be provided. When a spare tire assembly is specified (see 3.4.10.5), a spare wheel or rim shall be furnished.

* 3.4.10.5 Spare tire assembly. When specified (see 6.2), a spare tire assembly shall be furnished. The spare tire assembly shall include an inflated spare tire mounted on the spare wheel or rim. The spare tire shall be of the same size, tread design and load range (ply rating) as the tires furnished on the front axle of the vehicle.

3.4.10.6 Tire chain clearance. Tire chain clearance in accordance with SAE J683 shall be provided. Allowance for spring deflection shall be included.

* 3.4.11 Brakes. The brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.43 and 393.45 through 393.52. Brake linings shall be of nonasbestos material.

3.4.11.1 Power hydraulic type. Unless otherwise specified, class A, B and C vehicles shall be equipped with power assisted, hydraulic actuated, four-wheel service brakes.

3.4.11.2 Full air brakes. When specified for class A, B and C vehicles (see 6.2), for all class D, E and F vehicles, and when a trailer air brake system is specified (see 3.1.1.8), full air-application brakes shall be furnished on all wheels. The braking system, complete with all necessary components, shall include:

(a) Air compressor, unloader-head type, engine driven and engine lubricated, air or water cooled, and having a capacity of not less than 205 L/min (7.25 cubic feet per minute (cfm)) for class A, B, C, D and E vehicles, and not less than 340 L/min (12 cfm) for class F vehicles
(b) Air storage reservoir(s), each tank equipped with drain, and with safety and check valves between compressor and last reservoir tank
(c) Foot control, suspended or treadle type

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(d) Air control valves
(e) Air pressure gage, visible to the driver
(f) Low air pressure warning, visible and audible
(g) Service brake stop lamp switch
(h) Automatic moisture ejector
* (i) Automatic slack adjusters on cam type brakes or internal self-adjusting brakes on wedge and disc type brakes (model 4x2 and 6x4 only).

* 3.4.11.2.1 Air dryer. A replaceable cartridge desiccant type air dryer shall be installed in the air brake system. The dryer shall have the capability of removing not less than 95 percent of the moisture in the air being dried. The dryer shall have a pre-cooler and a filter to screen out oil and solid contaminants. The dryer shall have an automatic self-cleaning cycle and a thermostatically controlled heater to prevent icing of the purge valve.

3.4.11.3 Trailer brake control system. When a trailer air brake system is specified (see 3.1.1.9), a trailer brake control system shall be furnished in addition to the components specified in 3.4.11.2. The trailer brake control system shall include:

(a) Identification of emergency and service lines.
(b) Coincident control of trailer brakes with prime mover foot control.
(c) Independent hand control for trailer brakes.
(d) Towing vehicle protection valve with dash control and automatic break-away feature.
(e) Trailer stoplight control operable with foot brake and with hand control for trailer brakes.
(f) Air connectors for trailers with SAE J318 gladhand couplers mounted at the rear of the vehicle, located to prevent interference with trailer and easily accessible to the operator.
(g) Two SAE J844 pre-coiled air hoses, each not less than 2800 mm (110 inches) long when fully extended, with SAE J318 gladhand couplers on both ends of the hoses. The hoses shall be packaged and stowed in the vehicle for shipment.
(h) Dummy gladhand couplers with chains or cables.
(i) Prime mover only parking brake valve to permit prime mover parking brakes to be applied while charging the trailer air brake system.

3.4.11.4 Brake controls for use from a towing vehicle. When specified for vehicles with air brakes (see 6.2), the vehicle shall be furnished with a system for controlling the brakes from a towing vehicle (wrecker). The installation shall be complete with air brake couplers, a relay emergency valve with no-bleed-back feature (except when a spring applied emergency brake is furnished), additional air lines, and fittings. The service and

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emergency couplers shall be mounted on the front in a protected position providing for ready attachment of air hoses from a towing vehicle. The service and emergency couplers shall be identified and provided with dummy gladhand couplers with chains. The system shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard.

* 3.4.12 Cab. Unless otherwise specified, the chassis manufacturer shall furnish any type of his standard or optional non-tilting full width cab. When specified (see 6.2), a cab with a forward tilting hood and fender assembly, including tilting and locking mechanism, shall be furnished. Tilting shall not interfere with present or future installations of additional equipment, such as a snowplow or a front mounted winch. Cab doors shall be equipped with locks, operable from inside the cab through mechanical linkage, with both doors equipped with an external, key-operated locks. Drip rails shall be installed above the cab doors. When the front tires extend beyond the cab fenders, rubber fender extensions extending at least to the outside of the tire tread shall be furnished.

* 3.4.12.1 Cab interior. Unless otherwise specified, the cab shall have an upholstered, full-width, adjustable seat and back or individual, adjustable driver's seat and individual passenger seat. When specified (see 6.2), an individual, adjustable driver's seat and an individual passenger seat shall be provided. When specified for vehicles with air brakes (see 6.2), the driver's seat shall be the manufacturer's standard air ride suspension type. The color of the upholstery and the interior finish shall be compatible with the exterior color (see 3.1.1.1). White upholstery shall not be furnished. Interior lighting shall be provided. Two pairs of seat belts shall be installed. Outboard seats shall have combination pelvic and upper torso restraint seat belts.

* 3.4.12.2 Crew cab. When specified (see 6.2), a four-door, full-width crew cab shall be furnished in lieu of the standard full-width cab. The cab shall be equipped with two upholstered full-width seats and backs. The front seat shall be adjustable. Fastenings and three pairs of seat belts shall be installed for both front and rear seats. Front outboard front seats shall have a combination pelvic and upper torso restraint seat belts. Cab doors shall be equipped with locks operable from inside the cab through mechanical linkage, with both front doors equipped with an external key-operated lock. Cab doors shall have windows with crank-operated window regulators. Interior lighting shall be provided. The cab roof shall be of one-piece construction; or, if welded, the roof shall give the appearance of one piece, with weld seams being continuous, waterproof, and free of visible bumps or protrusions. Full length drip moldings shall be mounted above the doors. When a crew cab is furnished, the cab-axle dimension shall be measured from the back face of the crew cab.

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3.4.12.2.1 Seat spacing. With the front seat adjusted to the extreme forward position, there shall be not less than 7660 mm (30 inches), measured in a horizontal plane, between the front of the rear seat-back and the rear of the front seat-back. A kick-space height of not less than 70 mm (2.75 inches) shall be maintained between the floor and the rear of the front seat in all positions of adjustment. The leg room and space forward of the front seat shall be equivalent to that provided ahead of the seat in a two-door standard cab.

3.4.13 Steering. Power steering shall be provided.

* 3.4.14 Windshield wipers and washers. The vehicle shall be equipped with dual windshield wipers and windshield washers. The windshield wipers shall be multi-speed type and operated by either air or electric motor(s). When specified (see 6.2), and if electric wipers are furnished, intermittent wipers shall be furnished.

3.4.15 Bumpers. Unless the bumper is an integral part of the vehicle cab, a channel type front bumper shall be provided, except when a front mounted winch is specified a pipe type front bumper may be furnished (see 3.4.23.2). The rear bumper on type II maintenance trucks shall conform to 3.5.2.1. Rear end protection of both types of vehicles shall conform to Federal Motor Carrier Safety Regulation 393.86.

* 3.4.16 Tool stowage. Stowage space of sufficient size to accommodate a vehicle jack, hand tools, antiskid chains (for outside tires on duals only) and emergency reflective triangles shall be furnished. Stowage space for these tools may be furnished in the cab or in the tool boxes or cabinets. When stowage space for these tools is located outside of the cab, tool boxes or cabinets, it shall be weatherproof and shall provide for locking with a padlock.

3.4.16.1 Tools. When specified (see 6.2), the vehicle shall be furnished with tools required for exchanging a mounted tire assembly with the spare assembly and shall include at least a hydraulic jack, jack handle and wheelnut wrench. The jack shall be of such closed height as to permit its location under the axle, or other satisfactory lift point, at any wheel with the tire flat. The jack, without blocking, shall be capable of raising any wheel of the loaded vehicle to a height adequate to permit removal and replacement of wheel and tire assembly.

3.4.17 Heater and defroster. A hot water heater shall be provided. The heater shall have fresh air intakes. Discharge outlets shall be provided to direct heated air to the floor and to defroster louvers. The heater shall be complete with blower and mounted controls convenient to the driver.

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3.4.18 Controls and operating mechanism. All controls and operating mechanisms shall be located for left-hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of the operator to and from the driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. Instruments shall be panel mounted.

3.4.19 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

(a) Key-operated starting switch
(b) Ammeter, charging indicator or voltmeter
(c) Fuel gage
* (d) Oil pressure gage or red indicator warning light
* (e) Engine coolant temperature gage or high coolant temperature or low coolant level red indicator warning light
* (f) High coolant temperature or low coolant level alarm buzzer
(g) Speedometer with recording odometer
(h) Ash receptacle
(i) Dual sunvisors
(i) Driver's compartment ventilator other than window
(j) Tachometer for vehicles with a power takeoff driven hydraulic system.

* 3.4.20 Rearview mirrors. Outside rearview mirrors shall be mounted on both sides of the cab. The mirrors shall have flat and convex areas. The flat portion shall have not less than 320 square centimeters (cm^2) (50 square inches) of reflective area. The convex portion shall have not less than 155 cm^2 (24 square inches) of reflective area and a radius of curvature of not less than 510 mm (20 inches). The convex portion shall be attached to the lower mirror supporting arm and shall not interfere with use of the flat mirror. The mirrors shall have not less than two supporting arms. When specified (see 6.2), the curbside flat mirror shall be of the motorized type, with remote control. The mirror motor shall provide not less than 60 degrees horizontal rotational viewing range. When specified (see 6.2), the flat mirrors shall be electrically heated. Mirror remote and heating controls shall be within reach of the seated driver.

3.4.21 Horn. The manufacturer's standard electric horn shall be furnished. When specified (see 6.2), in addition, an air-operated horn shall be furnished on vehicles equipped with an air brake system.

* 3.4.22 Engine hour meter. An engine hour meter having a totalizing mechanism of not less than 9,999 hours shall be furnished for the chassis engine to register accurately the number of hours of operating time. The

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meter shall be of rugged construction to ensure continuous trouble-free performance under severe operating conditions. The engine hour meter shall be mounted on the cab instrument panel or in the engine compartment in a readable location.

* 3.4.23 Front mounted winch. When specified (see 6.2), the vehicle shall be furnished with a winch assembly mounted on the front of the vehicle. The winch shall be powered by the aerial platform hydraulic system. The winch shall conform to SAE J706. The winch shall be of the single drum type for hoisting and negotiating purposes and shall have one forward and one reverse speed. Dual winch controls shall be furnished, one inside the cab and one at the front bumper. Winch controls shall be so located as to provide no interference with the entrance or exit of the driver. The winch shall have a rated single line pull capacity of not less than 9070 kg (20,000 pounds) on the bare drum. The winch shall be equipped with a hydraulic relief valve set to restrict the line pull capacity to 9990 kg (22,000 pounds). The winch line speed on the bare drum shall be between 4570 and 9140 mm (15 and 30 feet) per minute at an engine speed equal to 50 percent of engine governed speed. The winch shall be wound with not less than 56 m (185 feet) of 16 mm (0.625 inch) diameter, preformed 6x37 improved plow steel, wire strand core wire rope and shall be equipped with a full capacity end chain and hook. An integral, adjustable, automatic safety brake shall be provided. The winch shall be equipped with a 4-way roller guide assembly, and shall be equipped with a shaft extension on the curb side with a capstan drive extension (AT&T type).

3.4.23.1 Winch drum guard. A winch drum guard shall be furnished. The winch drum guard shall restrict the cable to the area between the drum flanges. The guard shall consist of not less than 6.4 mm (0.25 inch) vertical side plates conforming to the outside radius of the drum flanges. Six bars, 9.5 mm (0.375 inch) by 29 mm (1.25 inches), shall be welded to the vertical side plates. Three bars shall be located on the top and spaced equally on the top radius, and three bars shall be located on the bottom and spaced equally on the bottom radius. The vertical distance between the vertical side plates and the flanges shall be not more than one-half the specified cable diameter.

3.4.23.2 Front bumper. When a front mounted winch is specified, the bumper shall be mounted forward of the winch. The open area on either side of the winch shall be covered with a combination step plate and gravel guard. The bumper shall be either a channel or pipe type. When a pipe type front bumper is furnished, the nominal diameter shall be not less than 75 mm (3 inches), and shall have a wall thickness of not less than that specified in Schedule 40 of ASTM A120. The pipe type front bumper shall have half-round ball ends.

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* 3.4.23.3 Combination step plate and gravel guard. When a front mounted winch is specified, a combination step plate and gravel guard shall be furnished. The guard shall be fabricated of not less than 14 gage (1.897 mm) (0.0747 inch) steel tread plate exclusive of the projections. The step plate shall be secured to the front bumper and shall be not less than 6.4 mm (0.25 inch) or not more than 9.5 mm (0.375 inch) from the chassis sheet metal. Rubber flaps may be employed to achieve these dimensions and shall be furnished when required by tilting hood/fender vehicles for unrestricted tilting. The step plate shall be capable of supporting 1460 kg/m^2 (300 pounds per square foot). The step plate shall not deflect more than 3.2 mm (0.125 inch) under the loads imposed.

3.4.23.4 Winch reel. When specified (see 6.2), a reel for payout and take-up shall be provided when a front mounted winch, specified in 3.4.23, is furnished. The reel shall be designed to operate from the end of the extended winch shaft. Reel yokes shall be controlled by a handle at the end of the reel shaft. The spindle shall be fitted with a bayonet type socket for attachment to shaft. Reel spiders and yokes shall be made from heat treated aluminum. The inside diameter of coil shall be approximately 510 mm (20 inches); maximum outside diameter approximately 710 mm (28 inches); thickness, or width, of coil approximately 150 mm (6 inches) to 230 mm (9 inches.)

3.4.24 Air conditioning. When specified (see 6.2), the vehicle shall be equipped with the manufacturer's standard air conditioning system. The air conditioning system shall include tinted windshield and tinted glass.

3.4.25 Servicing and adjusting. Prior to acceptance of the vehicle by the Government inspector, the contractor shall service and adjust each vehicle and its mounted equipment for operational use including at least the following: alinement of lights; adjustment of engine and brake systems; filling and charging of batteries; alinement of front wheels; inflation of all tires; complete servicing of chassis, engine, running gear and mounted equipment with grades of lubricants or hydraulic fluids recommended for the ambient air temperature at the delivery point; servicing of cooling system in accordance with 3.4.25.1; and servicing of windshield washer reservoir with water and appropriate additives.

* 3.4.25.1 Engine coolant. The engine coolant shall be a solution of ethylene glycol antifreeze and water or propylene glycol antifreeze and water, in equal parts of antifreeze and water by volume. Ethylene glycol antifreeze shall conform to ASTM D 4985 with not more than 250 parts per million silicates. Propylene glycol antifreeze shall be in the specific formulation approved by the engine and cooling system manufacturers. When specified (see 6.2), the percentage of antifreeze in the cooling system shall be increased to provide protection against freezing down to -54°C (-65°F).

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3.5 Vehicle body types.

3.5.1 Type I (flatbed). Type I vehicles shall be equipped with a full length safety tread steel flatbed type platform body extending from behind the cab to the rear of the chassis frame. A front bulkhead shall be provided, attached to the platform body. The overall width of the platform body and bulkhead shall be not less than 2410 mm (95 inches). The turntable pedestal of the boom shall be located in the platform body area. The overall height of the vehicle with the boom stowed in travel position, with tires inflated to maximum working pressure, shall be the minimum practicable but in no case more than 3810 mm (12 feet, 6 inches).

3.5.1.1 Body frame. Body framing shall be a completely welded structure with members of minimum gage thickness specified in table VI for carbon steel; high tensile steel may be furnished in two gages lighter weight in accordance with US Standard gage sizes. There shall be not less than 7 crossmembers for class A, B, C and D vehicles, and not less than 13 crossmembers for class E and F vehicles, including ends, and stub crossmembers as required for proper spacing over axles. For class F, model 6x4 vehicles, not less than 5 additional crossmembers shall be provided in the area of the rear bogie. The additional crossmembers may be joined by welding to the normally located crossmembers in the rear bogie area. Crossmembers shall be of full-channel construction, reinforced by gusset plates or brackets at points of attachment to longitudinal sills. Crossmembers shall extend the full width of the vehicle but may be multiple sections, spliced in place. Contact edges of crossmembers with longitudinal sills, and contact edges of welded reinforcements shall be welded for not less than 50 percent of the edge length. Longitudinal sills shall be constructed of structural steel channels or formed channels. Formed channel sills shall be reinforced within the sill, at each crossmember of the body mounting point, with formed channel reinforcements.

TABLE VI. Framing gages.

	US Standard gage No.	Equivalent mm (inches)
Crossmembers	10	3.416 (0.1345)
Side and end rails	10	3.416 (0.1345)
Longitudinal sills	8	4.176 (0.1644)
Reinforcements	8	4.176 (0.1644)

3.5.1.2 Flooring. The platform body floor shall be not less than 9.5 mm (0.375 inch) thick, one or two piece diamond safety tread steel with additional lateral supports provided at the wheel wells. Two piece floors shall be spliced longitudinally and completely welded the full length of the splice.

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3.5.1.3 Front bulkhead. The front bulkhead shall be capable of withstanding a horizontal static load equal to half the payload capacity of the vehicle without permanent distortion.

3.5.1.4 Cab protector. A cab protector of framing covered with expanded metal sheet steel, of not less than 9-gage (3.797 mm) (0.1495 inches) before the sheet steel has been expanded, and not less than 2290 mm (90 inches) in width, shall be furnished. The protector shall provide protection from falling objects, and shall have sufficient area above the cab roof for vehicle cleaning and maintenance. The cab protector shall cover the cab roof (including crew cab, when applicable), fenders and hood of the vehicle.

3.5.1.5 Lights for type I vehicles. In addition to lighting specified in 3.4.2.4, the following lighting and wiring shall be furnished:

(a) Demountable projecting load light on the end of the boom, with service line and socket when end of boom extends beyond rear of vehicle. A receptacle for the light on the rear of the body shall be provided. The light shall be in accordance with Federal Motor Carrier Safety Regulation 393.18
(b) An amber rotating, vehicular warning light in accordance with MS 51317-4 shall be provided. The rotating light shall be mounted with reinforcement at point of attachment on or near the left front corner of the cab guard for 360 degree visibility. The rotating light shall be operated by means of a separate switch with red indicator light located on the instrument panel and marked to indicate the function of the switch.

3.5.1.6 Tool boxes. Underslung, weatherproof, locking tool boxes shall be provided on both sides of the vehicle, extending from the back of the cab to the rear wheels. The tool boxes shall be not less than 510 mm (20 inches) deep by 510 mm (20 inches) high. Side hinged double panel doors with recessed handles shall be provided. The tool boxes shall be secured to the vehicle frame members and shall be constructed of not less than 12 gage (2.657 mm) (0.1046 inch) sheet steel.

3.5.2 Type II (maintenance). Type II vehicles shall be equipped with a line maintenance and construction type body having minimum exterior dimensions specified in table VII. The turntable pedestal of the boom shall be located in the body loadspace. Cabinet space may be utilized to accommodate outriggers and access steps. Safety-tread steel plate shall be installed in the loadspace and on top of the cabinets. The overall height of the vehicle shall conform to 3.5.1.

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TABLE VIII. Body dimensions (minimum).

Body dimensions	Vehicle class		
	A	B,C&D	E&F
	mm (inches)	mm (inches)	mm (inches)
Rear-of-cab to rear-axle (trunnion) dimensions (inches)	1520 (60)	2108 (83)	3048 (120)
Overall width of body (inches)	1930 (76)	2159 (85)	2286 (90)
Loadspace width (exclusive of wheelwells) (inches)	1270 (50)	1270 (50)	1270 (50)
Cabinet space (each side including wheelwells, excluding bumper and platform extension):			
Length (inches) without crew cab	2692 (106)	3251 (128)	4877 (192)
Length (inches) with crew cab	1829 (72)	2286 (90)	3912 (154)
Height at front (inches)	965 (38)	1016 (40)	1143 (45)
Height at rear (inches)	914 (36)	965 (38)	1092 (43)
Width (inches)	366 (14)	381 (15)	381 (15)
Loading (without permanent deformation):			
Floor of usable load space, live load on any square foot (pounds)	1270 (500)	1270 (500)	1270 (500)
Cabinets and shelves, live load on any square foot (pounds)	2540 (100)	2540 (100)	2540 (100)

3.5.2.1 Bumper and platform extension. A combination bumper and body platform extension shall be provided, extending beyond the rear of the cabinets. Safety tread steel plate shall be provided on top of the platform extension.

3.5.2.2 Stirrup steps. Stirrup steps shall be furnished on each side of the combination bumper and platform extension. Grab handles shall be installed to assist personnel entering the body. A stirrup step and grab handle shall be furnished on the rear of both rear cabinets for access to the top of the cabinets.

3.5.2.3 Access steps. When the turntable pedestal is mounted so that a loadspace length of 3 feet or more is forward of the turntable, access steps and grab handles shall be furnished on the curbside of the body located between the back of the cab and the front of the turntable pedestal.

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* 3.5.2.4 Vise. A vise shall be provided. The vise shall have at least 3-inch hardened replaceable general material jaws and hardened replaceable pipe jaws. The vise shall be of the swivel type and shall be bolted to the right rear corner of the platform extension.

3.5.2.5 Cabinets. Cabinet backs, partitions, and shelves shall be constructed of not less than 20 gage (0.919 mm) (0.0359 inch), high tensile steel or 18 gage (1.214 mm) (0.0478 inch) carbon steel. Cabinet doors shall be double panel, reinforced construction of not less than 22 gage (0.759 mm) (0.0299 inch) sheet steel. Doors shall be weatherproof and constructed to maintain fit and alinement in service. Doors shall have continuous rod, or hem-type, heavy duty hinges, with noncorrosive hinge pins or sockets. Latches shall have recessed handles. Cylinder type locks, keyed alike, shall be installed in each door. Moving parts of latches and locks shall be constructed of, or plated with, corrosion-resistant material. Cabinet doors over 1120 mm (44 inches) in length shall have pillar bolts or two latches. Horizontal doors shall have corrosion-resistant stay chains and shall be reinforced for use as shelves capable of supporting a live load of not less than 120 kilograms on any square meter (25 pounds on any square foot). The cabinet arrangement shall be the manufacturer's standard and shall provide the following:

(a) Full height vertical cabinets equipped with bolted or screwed-on hooks;
(b) Horizontal cabinets equipped with small parts bins with adjustable spacers, shelves and partitions, and a rubber-goods compartment;
(c) Forward cabinets equipped with not less than two adjustable tray-type shelves and provisions for mounting a water cask (see 3.5.2.9) in a curbside compartment;
(d) Safety tread steel plate shall be provided on the top of each cabinet;
(e) A front bulkhead conforming to 3.5.1.3.

3.5.2.6 Cab protector for type II. A cab protector conforming to 3.5.1.4 shall be provided.

3.5.2.7 Lights for type II vehicles. In addition to the lighting specified in 3.4.2.4, type II vehicles shall be provided with the lights specified in 3.5.1.6(a) and (b) and the following:

(a) Lights with toggle switches and guards in each cabinet compartment;
(b) Lights on universal mounts of sufficient candlepower to illuminate the body loadspace, all equipment contained therein, and all controls;
(c) Two, 4.5 inch minimum diameter sealed beam floodlights, with universal mounts, one on each side of the top of the cab protector, with controls on the outside and operable by an operator from inside the cab through an open window for illuminating the work area;

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(d) Wiring for the body in accordance with the applicable portion of the General Section, SAE J1292. All wiring exposed to possible physical damage, such as wiring on a compartment floor, or wiring where tools or materials may cause the wire to be frayed or shorted, shall be protected by the use of heavy-wall, thermoplastic conduit or steel conduit having waterproof connections. Grommets shall be used for all wiring going through any hole or body frame work.

3.5.2.8 Water cask. When specified (see 6.2), a standard commercial 11.4 L (3-gallon) insulated potable water cask shall be furnished. The inside liner shall be stainless steel or a nontoxic poly material. The outside shell shall be galvanized steel or equivalent. The throat area shall be of sufficient size to allow cleaning the interior of the water cask. A nylon or brass chrome plated spigot, flush mounted to prevent damage, shall be furnished. Means shall be provided to drain excess water to the ground. A cup dispenser for dispensing 0.18 L (6 ounce) hot vending cups shall be furnished. Lugs and bracket(s) shall be provided for mounting the cask in a vertical compartment on the curbside of the vehicle. Means shall be provided to retain the water cask and to minimize vibration.

3.5.3 Turntable pedestal and outrigger mounting, types I and II. No part of the body (cabinets, floor, turntable pedestal, outriggers or subframe) shall be welded to the chassis frame. Size 28, 32 and 35 vehicles shall be mounted in accordance with the manufacturer's standard practice. For size 45, 50 and 60 vehicles, the turntable shall be attached to a pedestal mounted on a full length subframe extending from immediately behind the truck cab to the end of the chassis frame rails. The full length subframe shall be secured by brackets bolted to the chassis frame rails. The outriggers for all sizes shall be attached to the turntable pedestal or to a subframe.

3.5.4 Floor and cabinet mounting, types I and II. For type I, in all sizes, the mounting of the safety tread steel plate platform body shall incorporate U-bolts, twin studs or brackets, and breaker strips and shall be in accordance with 3.5.4.2 through 3.5.4.4. For type II, the mounting of the cabinets and the loadspace floor shall conform to 3.5.4.1 through 3.5.4.4 or to 3.5.4.5, at the manufacturer's option.

3.5.4.1 Flexible mounting for cabinet-floor assembly. The cabinets and the loadspace floor on type II shall be an integral assembly, secured independently of the turntable and the outriggers to the chassis frame or turntable subframe with U-bolts, twin studs or brackets and shall include a wood breaker strip. All but the rearmost cabinet-floor assembly mountings shall be spring loaded to provide for movement of the cabinet-floor assembly independent of the subframe, chassis frame, turntable and outriggers.

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3.5.4.2 U-bolts or twin studs. When U-bolts or twin studs are used for mounting the floor or cabinet-floor assembly, there shall be not less than four U-bolts or twin studs per side having a 14 mm (0.563 inch) minimum body diameter with a 9.5 mm (0.375 inch) minimum thread diameter. Tie plates shall be at least 13 mm (0.5 inch) thick. The vehicle chassis frame and the full length subframe shall be braced, using wood blocks, at each mounting point unless the mounting point is located at a full depth crossmember. Blocks shall incorporate a keeper strap or groove for the mounting bolt and shall be of a width to assure retention. Two tieback straps shall be provided, one bolted to each side of the rear, to maintain the floor alinement on type I or cabinet-floor assembly alinement on type II.

3.5.4.3 Brackets. When brackets are used, they shall be bolted to the web of the full length subframe or of the chassis frame rails. The body mounting brackets shall provide means of drawing down the body on the subframe or chassis rails and provisions shall be made to prevent lateral shifting of the breaker strips. When additional holes are required to secure the mounting brackets to the chassis frame rails, they shall be located within the area of the rail which is designated as being safe for drilling in accordance with the chassis manufacturer's body builders layouts. Attachments shall not interfere with nor obstruct existing chassis components.

3.5.4.4 Breaker strips. A hardwood or dense southern yellow pine breaker strip not less than 27 mm (1-1/16 inches) in finished thickness shall be installed between the longitudinal sills of the floor or cabinet-floor assembly and the full length subframe. Breaker strips shall have a taper of 25 mm (1 inch) in 460 mm (18 inches) at the forward end.

* 3.5.4.5 Rigid mounting for cabinet-floor assembly. The cabinets shall be bolted and the loadspace floor shall be bolted or welded to the full length subframe. The full length subframe and the chassis frame shall be tied together in such a manner as to act as a single member resisting bending and distortion of the cabinets. The section modulus of the full length subframe and the chassis frame combined shall be not less than 1150 cubic centimeters (70 inches cubed) per frame rail. A wood breaker strip conforming to 3.5.4.4 shall be provided when the cabinets and the loadspace floor are an integral assembly. When the cabinets are mounted independently of the loadspace floor, wood or metal breaker strips shall be provided when needed to provide relief at rivet heads, bolt heads, or other subframe irregularities.

* 3.6 Aerial tower. Aerial tower structural components shall have a calculated design stress not greater than that allowed by ANSI A92.2. The vehicle shall be designed and constructed in conformance with ANSI A92.2.

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3.6.1 Turntable. The turntable assembly shall be rotated by a hydraulically operated motor. The turntable shall be constructed to eliminate all side thrusts on the motor shaft.

3.6.2 Fiberglass reinforced plastic. The fiberglass boom, baskets and components shall consist of extruded, molded or spiral-wound fiberglass. Each component shall conform to the performance requirements for electrical insulation specified herein. All components shall be made from a combination and multiple layers of sterile electrical grade fiberglass roving cloth, mat or strands with high impact strength resins with weathering and fire retardant characteristics employed. The insulated boom shall have a hard, smooth surface inside and outside to prevent contamination and to preserve dielectric characteristics. The exterior surface shall be of uniform thickness. When gel-coat is applied, thickness of gel-coat shall be not less than 0.15 mm (0.006 inch) and not more than 0.76 mm (0.030 inch). The parts shall be free from uncured areas, gaps, cracks, holes, blisters, fuzzy surfaces, resin pockets, areas lacking resin, laps, wrinkles, delamination, air or gas pockets, patches, porosity and other similar defects. The material shall be essentially void free. The entire surface shall be smooth, fully cured and free of tackiness. Percentage of glass by weight shall be approximately uniform. The plastic construction insulation shall meet or exceed the requirements of 3.6.3.

* 3.6.3 Dielectric properties. The complete aerial platform (aerial device) shall conform to and shall pass the tests required for a category C aerial device rated at 46 kV in accordance with ANSI A92.2, without exceeding the maximum allowable boom current specified in ANSI A92.2 for a 46 kV rating. Any mechanism within or along the upper boom shall be made of insulating materials, or so insulated as to demonstrate the same dielectric properties as the boom.

3.6.3.1 Basket liner dielectric properties. A decal stating "NO INSULATION VALUE WITHOUT LINER" shall be applied on each basket. The liner shall withstand a minimum of 50,000 volts ac, 60 Hz, without evidence of dielectric puncture.

* 3.6.4 Boom assembly. The boom assembly shall be of the telescopic type and shall be hydraulically operated. The boom assembly shall be attached through a rotating turntable to a pedestal utilizing self-alining bearings and shaft or shall be machined to assure alignment of the shaft and bearing surfaces. Under dynamic conditions, the boom assembly shall support the specified load placed in the aerial platform in any extended position without permanent deformation of any component in or along the boom or supporting structure.

3.6.4.1 Boom construction. The boom shall be designed, engineered and constructed of materials that display the minimum weight conducive to the configuration and maximum performance requirements specified herein.

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Hydraulic lines and other associated parts which are fastened to or inside the boom shall be positioned for easy inspection, ready maintenance or possible replacement and shall have maximum protection against damage afforded by the configuration of the boom. The boom shall be attached to the aerial platform and to the turntable by replaceable shafts. Rotating bearing surfaces of the boom at the turntable which are subjected to continuous wear shall be provided with replaceable bearings or bushings. Lubrication fittings or permanently lubricated components shall be provided on all rotating parts. Bearing surfaces shall be adequate to resist torsional moment when inherent with the design. When a completely closed boom is furnished, means shall be provided within the boom to absorb moisture. The boom shall be provided with inspection openings at critical areas.

3.6.4.2 Boom operation. The rotating turntable and boom assembly shall be hydraulically actuated by separate mechanisms and shall operate independently or in any combination of raising, lowering, and rotating at the same time. When in combination, hand metering of the controls may be necessary. The boom operating mechanism shall be capable of limiting the lifting capabilities from rest position, within safe limits of design stress to prevent overloading. For testing purposes, means shall be provided in the hydraulic system to perform the overload tests. The vehicle shall be provided with a device or shall be otherwise designed to prevent the boom from reaching a position that would cause the vehicle to be unstable.

3.6.4.3 Boom stowage. For normal stowage, travel and shipment of the vehicle, the boom shall retract to an essentially horizontal position in a vertical plane parallel to the longitudinal centerline of the vehicle. Means shall be provided to hold down the boom assembly and basket or aerial platform to prevent unactuated raising of the boom when in transit. Saddles and a tiedown device shall be provided to support the boom and basket or aerial platform assembly and to prevent vertical and side movement in transit. Boom saddles shall be sufficiently padded to prevent damage to the boom. The supports shall not interfere with the entrance and exit of personnel to the body loadspace.

3.6.5 Aerial basket. Unless a guardrail type aerial platform is specified (see 3.6.5.2), a one-man fiberglass basket shall be furnished and shall have a load capacity of not less than 1350 N (300 pounds). The minimum interior dimensions of the basket shall be 1020 mm (40 inches) depth to floor, 546 mm (21.5 inches) wide, and 546 mm (21.5 inches) front to back. Minimum interior dimensions shall be measured at the bottom of the basket just above the bottom radii. A non-rectangular basket may be furnished, provided the total usable floor area is equivalent to that of a rectangular basket with the minimum dimensions specified. The basket shall be fully enclosed on five sides. The basket shall have a molded-in or bonded on-step, protruding toward the outside, to aid entrance and exit of the operator. The fiberglass basket shall be provided with a guard to protect the controls.

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3.6.5.1 Basket with liner. A single-shell fiberglass basket shall be provided, with liner. Dielectric strength performance requirements of the basket and liner shall be as specified in 3.6.3.2. The liner shall enclose the basket interior. The liner shall be easily removable from the basket for cleaning and testing the dielectric capacity. The floor of the liner shall be provided with a removable, nonconductive scuff mat.

3.6.5.1.2 Basket cover. A waterproof basket cover and means to retain the cover on the basket, while in stowage or stowage-in-transit position, shall be provided on the basket and also may enclose the controls; otherwise, a cover shall be provided to completely enclose all control mechanisms. Provisions shall be made to store the cover(s) in one of the tool boxes or compartments of the body.

3.6.5.2 Guardrail type aerial platform. When specified for size 60 vehicles (see 6.2), an uninsulated guardrail type aerial platform shall be furnished. The platform shall have a load capacity of not less than 340 kg (750 pounds). Two guardrails and a kickrail shall be provided around the perimeter of the platform. The kickrail shall be not less than 100 mm (4 inches) in height. The lower and upper guardrails shall be not less than 510 mm (20 inches) and 1070 mm (42 inches) respectively from the floor of the platform. The flooring shall be of the open grating type. Interior dimensions shall be not less than 1070 mm (42 inches) in depth, 1020 mm (40 inches) in width and 760 mm (30 inches) front to back. A gate with a positive safety latch to hold the gateway closed shall be provided. Protruding edges shall be padded with a resilient replaceable bumper. The guardrails shall be constructed of aluminum alloy.

3.6.5.3 Flooring. The basket or aerial platform shall have a reinforced floor. No point in the flooring shall deflect more than 6.4 mm (0.25 inch) nor be permanently deformed when subjected to a vertical load of 1350 N (300 pounds) covering an area of 160 cm^2 (25 square inches).

* 3.6.5.4 Safety belts. Lineman type safety belts with positive-lock quick release features shall be provided. Each belt shall have a design strength of 910 kg (2,000 pounds). The safety belts shall be designed and positioned to allow freedom of movement of the workman in the basket. Means shall be provided to attach fasteners to the end of the boom and to prevent the fastener attachment from being pulled through the plastic wall, in event of leveling system failure. Fasteners and attachments shall be corrosion-resistant. A safety belt shall be provided for each personnel space available in the basket or aerial platform.

3.6.5.5 Tool tray. A fiberglass tool tray shall be provided to fit over the edge of the basket or aerial platform guardrail. Provisions shall be made to secure and stow the tray in a tool box or compartment of the body.

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* 3.6.5.6 Basket engine starter and cutoff. An engine starter and cutoff for the truck engine, remotely operated from the basket or aerial platform shall be furnished. The engine starter and cutoff shall not reduce the insulating characteristics specified in 3.6.3.

3.6.5.7 Basket leveling system. A hydraulic leveling system shall be provided for keeping the basket or aerial platform horizontal in all boom positions, with rated load positioned on the basket or aerial platform floor. Provisions for adjusting the level of the basket or aerial platform shall be provided. The leveling system components shall be provided with maximum protection based on design features and safety. Provisions to prevent tipping of the basket or aerial platform in case of failure shall be provided. The leveling system shall be automatically operated in all extended positions that the boom is capable of attaining. The aerial basket shall be capable of being released and easily tilted over, by one man, near ground level, to remove a disabled operator or to dump out water.

* 3.6.6 Controls. A single lever basket control and a complete set of hand-operated controls for maneuvering the basket or aerial platform shall be provided at the basket or aerial platform and at the base of the boom, respectively. The placement of the controls shall prevent inadvertent actuation by the operator. Controls for maneuvering the basket or aerial platform shall be located on or at the right or left side of the basket or aerial platform in a protected position easily accessible to the operator facing the work area. Controls mounted on the front or rear of the basket or aerial platform shall not be provided. Smooth operation, with no lurching of the boom assembly, shall be possible. The controls shall be self-centering and shall revert to a neutral position when released by the operator. All boom and outrigger controls shall be the dead-man type operable with one hand but whereby two separate actions must be taken to actuate the control. The controls at the base of the boom shall be easily accessible from the floor of the body at the base of the boom. Their operation shall be capable of overriding the controls at the basket or aerial platform. A safety interlock system shall be provided to prevent operation of the boom when the outriggers are not extended. Ground controls shall be placed to minimize possibility of use by unauthorized personnel. A means shall be provided to rotate the turntable; retract, lower and stow the boom; retract the outriggers and place the vehicle in travel position in case of power source failure. The switches, instruments and pressure gages on the lower control station shall be panel mounted, identified as to their function and installed in a manner to facilitate removal and service. Gages and instruments shall be shockproof, waterproof and weatherproof. Identification markings shall be on a photoetched, corrosion-resistant plate.

* 3.6.7 Outriggers. Two hydraulically operated outriggers shall be provided if the minimum attainable height from the ground to the bottom of the basket or aerial platform is less than 11 m (36.5 feet). Four

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hydraulically operated outriggers shall be provided if the maximum attainable height is 11 m (36.5 feet) or greater. Outriggers shall ensure adequate stability with relation to ground while operating the basket or aerial platform under maximum loads. Outrigger cylinder hydraulic fittings, hoses, tubing and pilot check valves shall be protected by metal guards, but shall be accessible for service. Outriggers shall be designed so that they may be extended over curbings and sidewalks when operating in narrow areas. Individual control for each outrigger shall be furnished to permit leveling of the vehicle on inclined surfaces. Controls for the left outrigger(s) shall be provided at the rear left corner of the vehicle and controls for the right outrigger(s) shall be provided at the rear right corner of the vehicle. An automatic pilot check valve shall be provided with an automatic, positive spring-actuated lock, or an automatic pilot check valve shall be provided which holds the outrigger in the retracted position without settling and which may be manually released. Interlocks shall be furnished to prevent operation of the telescopic boom assembly until the outriggers have properly descended.

3.6.7.1 Outrigger pads. Each outrigger shall be provided with a hinged type ground pad.

3.6.7.2 Outrigger warning light. A red warning light shall be installed in the cab in a conspicuous location to warn the driver when the outriggers are extended.

3.6.7.3 Circular level. Two circular levels, (Geir and Blume or equal), with 0 to 5 degrees reading shall be furnished. Circular levels shall be mounted in a protected position, visible and readable from the outrigger controls. Circular levels shall indicate the degree of body level as the outriggers are maneuvered. Circular levels shall be resistant to inclement weather and shall be corrosion-resistant. A warning plate (see 3.1.1.2.1 and 3.1.1.2.2) shall be mounted adjacent to each level to indicate in degrees the maximum longitudinal incline and the transverse slope on which the vehicle is capable of operating.

3.6.8 Hydraulic system. A hydraulic fluid, nonconductive system consisting of all components required to perform all hydraulic operations of the boom assembly and outriggers with the degree of control specified herein shall be provided. The hydraulic system shall also include the components required to power the front mounted winch, when furnished (see 3.4.23). Mechanisms and reservoirs shall be positioned or shielded to prevent hazardous fluid contact with heated surfaces or electrical apparatus. An automatic throttle actuator shall be furnished which increases the speed of the auxiliary or truck engine when the pump is being loaded. An accumulator may be used as a component of the hydraulic system to provide the maximum degree of temperature control under continuous service. All critical hydraulic components shall have a minimum bursting strength of at least four times the operating pressure for which the system is designed.

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3.6.8.1 Hydraulic cylinders. All hydraulic cylinders shall incorporate seals to prevent entrance of moisture and dirt into the cylinders and leaks of hydraulic fluid from the cylinders. Preformed packings shall be used in the cylinders where static seals are required. Manufacturer's standard seals shall be furnished where applied to reciprocating parts. All hydraulic cylinders for actuating the boom and outriggers shall be provided with pilot check or holding valves to prevent descent of the boom assembly and retraction of the outriggers in case of loss of hydraulic pressure. Means shall be provided to install pressure gages for testing hydraulic system pressure and relief settings. All surfaces of hydraulic ram piston rods shall be chrome plated with sufficient thickness to withstand a 30-hour salt spray test in accordance with ASTM B117.

3.6.8.2 Hydraulic lines and fittings. Hydraulic lines shall be located to facilitate maximum ease of removal and installation, or visual inspection for failures and maximum protection from damage during operation of the boom assembly. Flexible, hydraulic hose or collector rings shall be used where flexing of hydraulic lines is encountered and where installation of components requires a flexible coupling. Standard straight-thread, flared type fittings or adaptions to flared type fittings, or tapered pipe thread with dry seal fittings shall be furnished. Hydraulic lines and fittings shall be air and liquid tight and shall be installed to prevent vibration and chafing. The system shall be designed to prevent hydraulic transients. Provisions shall be made for bleeding air from the hydraulic system and for completely draining all hydraulic lines. A manual type bleeder shall also be provided as on-vehicle-equipment on closed-center, non-self bleeding systems. Hose assemblies and all seamless steel hydraulic lines shall be adequate to provide the rate of flow, temperature and pressures required to operate the system smoothly and efficiently. Galvanized pipe and copper tubing shall not be acceptable.

* 3.6.8.3 Hydraulic pump. A hydraulic pump with necessary pressure and flow capacities to operate all hydraulic powered components shall be provided. The pumping system shall include an adjustable pressure relief valve or unloading valve with a range of adjustment which allows raising or lowering, extending or retracting of the boom and basket with any working or test load specified herein. A pressure gage shall be provided on the discharge side of the hydraulic pump.

* 3.6.8.4 Pump drive system. The hydraulic pump shall be driven by a heavy duty power takeoff (PTO) unit attached to the truck chassis transmission. For emergency use, except when a rechargeable electrical or auxiliary engine driven system is specified, an auxiliary hydraulic pump, driven by an electric motor powered from the vehicle chassis electrical system, shall be provided to rotate the turntable, retract the outriggers, lower the booms and place the booms in stowed position in the event of primary power source failure. The system shall be equipped with cut-in devices and controls at the basket(s) and at the lower control station. The

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system shall provide adequate pressure and flow capacities for operation when the primary power source fails. When specified, in addition to the PTO, the hydraulic pump shall be driven by a rechargeable electrical system (see 3.6.9), or by an auxiliary engine (see 3.6.10). Means of rapid conversion from one power source to the other, from within the aerial basket and from the ground, shall be provided. Chain drives shall not be used in any system.

3.6.8.5 Hydraulic fluid reservoir. A hydraulic fluid reservoir shall be provided, with sufficient capacity to hold the required volume of hydraulic fluid, with not less than 10 percent for expansion space. The reservoir shall be adequately baffled. The reservoir shall be provided with a filler opening, a removable fill opening oil strainer and a vented filler cap, accessibly located to facilitate filling the tank from a 3.79 L (one-gallon) container using a funnel. The vented filler cap shall have means to prevent fluid from splashing out of tank during movement of vehicle. The filler cap shall be attached to the tank by a corrosion-resistant chain or cable to prevent loss. Provision shall be made for the inspection and cleaning of the interior and for draining the reservoir. The reservoir drain shall be located so that the fluid will not drain on other components and shall allow recovery of the fluid in a container. The reservoir shall have a photoetched, corrosion- resistant plate to indicate type of hydraulic fluid furnished and the maximum fluid level in the reservoir with the boom retracted for travel position and the hydraulic lines and components full of fluid. A sight level or dip stick gage shall be furnished to observe level of fluid. The tank shall be clean. A plate (see 3.1.1.2.2) shall be secured to the hydraulic fluid reservoir stating "USE ONLY RECOMMENDED TYPE OF HYDRAULIC FLUID."

3.6.8.5.1 Hydraulic fluid. Unless otherwise specified, hydraulic fluid conforming to the manufacturer's standard or recommended requirements shall be provided in the hydraulic system. When specified (see 6.2), hydraulic fluid conforming to MIL-H-83282 shall be provided in the hydraulic system.

* 3.6.8.6 Oil filter and strainers. A full flow type, 10-micron discharge oil filter in the return line to the reservoir or in the pressure line immediately after the pump shall be provided. A 100-mesh suction line strainer shall be provided on the suction side of the hydraulic system. Filter and suction strainer shall be easily accessible for periodic cleaning or for replacement of the filter cartridge. Filter and strainer shall have sufficient capacity to allow maximum rated pump discharge.

3.6.8.7 Hydraulic fluid temperature. The hydraulic fluid temperature shall be maintained within safe limits to prevent seal damage or approach to the flash point of the fluid during all operating conditions of the vehicle and use of the aerial tower hydraulic system for long periods of time. Excessive heating of the fluid (more than 13°C (55°F) above ambient in an ambient of 52°C (125°F) or equivalent at lower ambients) shall not be

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acceptable under any operating condition. Provisions shall be made to prevent malfunction or unactuated operation of the hydraulic control system due to changes in fluid temperature.

* 3.6.9 Rechargeable electrical system. When specified (see 6.2), a rechargeable electrical system shall be provided to drive the hydraulic pump, in addition to the normal power takeoff drive. An electric motor shall be mounted in conjunction with the hydraulic pump. Mounting shall minimize transfer of vibration. The electrical system shall include:

(a) A minimum 12-volt dc motor
(b) Industrial battery(s) of not less than 450 ampere-hour capacity
(c) Provisions for charging battery(s) from vehicle alternator (see 3.4.2.3)
* (d) A 220-volt, single phase battery charger with a minimum 30-ampere dc output
(e) Hour meter
(f) Battery charge indicator
(g) Pressure switches and control circuits
* (h) Battery isolator connected to the chassis charging system which prevents a drain of the chassis battery during boom operations

* 3.6.10 Auxiliary engine. When specified (see 6.2), an auxiliary air cooled engine shall be provided to drive the hydraulic pump assembly, in addition to the normal power takeoff drive for the hydraulic pump. When an air compressor is specified (see 3.6.11), an auxiliary engine shall be provided to drive both the hydraulic pump and the air compressor. The auxiliary engine shall operate on the same type of fuel as the main chassis engine. The engine shall be mounted on the body platform or loadspace. Mounting shall minimize transfer of vibration. The engine shall be capable of being operated from the basket or aerial platform. The auxiliary engine shall have sufficient output power to drive all operating components other than the vehicle. Gasoline engines shall be capable of operating on unleaded gasoline with a research octane rating of 91 without modification or adjustment.

3.6.10.1 Auxiliary engine components. The auxiliary engine shall operate on the same type of fuel as the main chassis engine and shall be a two- or four-stroke diesel or a four-stroke cycle gasoline engine equipped with at least the following components:

(a) Magneto-type ignition (gasoline engine only)
(b) Starter, 12-volt
(c) Resistor-type spark plugs (gasoline engine only)
(d) Air cleaner
(e) Automatic choke (gasoline engine only)
(f) Power plant heaters (see 3.4.1.6)

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(g) When specified (see 6.2), an engine hour meter for auxiliary engine having a totalizing mechanism of not less than 9,999 hours shall be provided.

3.6.10.2 Auxiliary engine control panel. The engine throttle, key-operated starting (ignition) switch and palm button starter shall be installed in a weather- and waterproof cabinet on a control panel, and each shall be identified as to its function. The cabinet shall be provided with a side hinged cover with a spring type latch and provisions for locking with a one-inch padlock.

3.6.10.3 Electrical system. The auxiliary engine electrical system shall utilize the truck-chassis battery and shall be in accordance with Federal Motor Carrier Safety Regulations 393.30, 393.31 and 393.33.

3.6.10.4 Auxiliary engine fuel pump. A fuel pump shall be furnished on the auxiliary engine. When lifting fuel against a head of 1220 mm (four feet), the pump shall have sufficient capacity to supply the engine at its maximum rated load. An in-line fuel filter shall be provided.

3.6.10.5 Governor. A variable-speed governor shall be furnished. The governor shall automatically control the auxiliary engine speed at the selected rated speed at all fluid pressures and flows required. A manually operated hand throttle shall control the engine at all speeds from the manufacturer's recommended minimum governed speed up to the maximum operating speed.

3.6.10.6 Auxiliary engine fuel supply. Unless an auxiliary engine fuel tank is specified (see 3.6.10.6.1), the auxiliary engine shall utilize the vehicle's main engine fuel tank(s). An accessible shut-off valve shall be provided in the fuel line. Fuel line(s) shall be in accordance with Federal Motor Carrier Safety Regulations 393.65 and 393.67.

3.6.10.6.1 Auxiliary engine fuel tank. When specified (see 6.2), an auxiliary engine fuel tank shall be furnished. The fuel tank shall have a capacity to provide fuel for at least 8 hours of full-load operation. The fuel tank cap shall be attached to the tank by means of a corrosion-resistant chain or cable. A sump shall be provided in the fuel tank suitable for collection and accumulation of water. A drain shall be located at the bottom of the sump to allow complete drainage of the fuel tank and sump. The drain shall be located to prevent spillage of fuel on any area of the vehicle. The tank filler shall be provided with a protective screen and vent. A shut-off valve shall be provided in the fuel line adjacent to the fuel tank. A corrosion-resistant plate shall be furnished and marked denoting the type of fuel to be used and shall be located in a conspicuous location near the filler cap.

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3.6.10.7 Auxiliary engine exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83 and shall be furnished with a spark arrester with not less than an 80 percent efficiency rating when rated in accordance with SAE J350. Means shall be provided to vent exhaust fumes away from the cab and body. The exhaust system or hydraulic system shall be shielded so that, in the event of a hydraulic line rupture or hydraulic fluid leak, the hydraulic fluid will not come into contact with any part of the exhaust system.

3.6.11 Air compressor. When specified (see 6.2), an air compressor shall be furnished. The air compressor shall be powered by the auxiliary engine (see 3.6.10). The compressor shall be the two-stage type with constant speed control, shall have an actual displacement of not less than 0.85 cubic meter per minute (30 cubic feet per minute), and shall provide a pressure of not less than 1.05 megaPascals (MPa) (150 pounds per square inch (psi). The air compressor shall be complete with receiver, moisture separator, automatic oilers, shock and waterproof pressure gage, safety valve and all necessary equipment for satisfactory operation of the compressor. The compressor shall be installed to minimize transfer of vibration. The receiver tank shall have a capacity of not less than 114 L (30 gallons). An air pressure regulator having a setting range of at least .34 MPa (50 psi) to 1.05 MPa (150 psi) shall be furnished. An air line extending from the compressor to a protected outlet at the curbside of vehicle shall be provided. An air line extending from the compressor to an outlet at the basket or aerial platform shall be provided and shall be of the nonconductive type. An air line shut-off valve shall be provided at the basket or aerial platform outlet.

3.6.12 Hydraulic tools.

3.6.12.1 Hydraulic ground tool outlet. Unless otherwise specified (see 6.2), a hydraulic tool outlet shall be provided in a curbside cabinet or in a protected position on the rear or under the platform. Discharge port(s) and return port(s) shall be provided with a corrosion resistant cap, secured with a corrosion-resistant chain, to protect the couplings and the valve from contamination. The tool outlet shall use the hydraulic fluid pressure from the hydraulic system of the aerial platform. One tool circuit with quick disconnect, dripless couplers shall be furnished for attaching the pressure and return hose lines for motor driven hydraulic tools. The oil flow for the tools shall be regulated by control valve(s) to meet the flow requirements of the hydraulic tools. The high system pressure (working pressure) shall be not less than 11.0 MPa (1,600 psi) and the back pressure in the return line shall not exceed 1.05 MPa (150 psi).

3.6.12.1.1 Basket hydraulic tool outlet and hose. Unless otherwise specified (see 6.2), a hydraulic tool outlet shall be furnished in the basket or at the aerial platform. An 2440 mm (8-foot) dual hydraulic hose, for use in basket or aerial platform, shall be furnished. The hose shall be a

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rubber, nonconductive type with quick disconnect, dripless fittings on each end. Hose fittings shall be compatible with the basket or aerial platform outlets and with the tools specified in 3.6.12.3 through 3.6.12.8. The hose ends shall be provided with dust caps.

3.6.12.2 Ground tool hose. Unless a hydraulic ground tool outlet is not required or unless a hydraulic hose reel is specified (see 3.6.11.11), two 15 m (50-foot) lengths of hydraulic wire braid hose with quick disconnect, dripless couplers on both ends for use at ground level shall be furnished. Warning plate markings (see 3.1.1.2.1) shall warn that the wire braid hydraulic hoses shall not be used on or near electrical lines. The hose shall be furnished with dust caps. Hose fittings shall be compatible to the tool outlet(s) specified in 3.6.12.1 and the tools specified in 3.6.12.3 through 3.6.12.8.

3.6.12.3 Hydraulic capstan hoist. When specified (see 6.2), a hydraulic capstan hoist shall be furnished. The drum shall be one direction drive and shall have a backstop on the output shaft to prevent reverse travel. (Lowering will be accomplished by letting the rope slip on the drum.) The capstan hoist shall be complete with a link chain and screw clamp attachment device for mounting to surfaces other than on a round pole. The hoist shall have a rated single line pull capacity of not less than 570 kg (1,250 pounds) on the bare drum. The hoist shall have a remote control, foot-operated, open-center valve.

3.6.12.4 Hydraulic impact wrench and drill. When specified (see 6.2), a hydraulic impact wrench and drill shall be furnished. This combination tool shall be reversible and capable of installing or removing lag screws and other hardware. The wrench shall have a 0.625 inch standard bolt size capacity with a 0.5 inch square drive. The tool shall be capable of developing not less than 270 Newton meters (N·m) (200 foot-pounds) of torque with at least 1,350 impacts per minute. A quick change drill chuck shall be furnished.

3.6.12.5 Hydraulic chain saw. When specified (see 6.2), a hydraulic chain saw shall be furnished. The saw shall have a chipper-type chain and shall be suitable for cutting trees up to 380 mm (15 inches) in diameter.

3.6.12.6 Hydraulic chain saw, pole mounted. When specified (see 6.2), a hydraulic chain saw, pole mounted, shall be furnished. The chain saw shall be light weight (approximately 4.5 kg (10 pounds)) and shall have a cutting capacity of 230 mm (9 inches). The pole shall be approximately 1520 mm (5 feet) long and shall have a fiberglass type of body to provide good insulation.

* 3.6.12.7 Hydraulic circular saw, pole mounted. When specified (see 6.2), a hydraulic circular saw, pole mounted, shall be furnished. The saw shall have a capacity of 95 mm (3.75 inches) and shall operate at

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approximately 2700 m (8900 feet) per minute. The pole shall be approximately 1520 mm (5 feet) long and shall have a fiberglass type of body to provide good insulation. The saw shall have at least a 40-tooth blade.

3.6.12.8 Hydraulic pruner, pole mounted. When specified (see 6.2), a hydraulic pruner, pole mounted, shall be furnished. The pruner shall have a capacity of 50 mm (2 inches) in diameter. It shall be mounted on a fiberglass type pole to provide good insulation and shall be approximately 2130 mm (7 feet) long.

3.6.12.9 Hydraulic fittings. Quick disconnect, dripless fittings shall be provided on each hydraulic tool. The fittings shall be compatible with the hoses specified in 3.6.12.1.1, 3.6.12.2 and 3.6.12.11.

3.6.12.10 Hydraulic tool stowage. Hydraulic tools, when furnished, shall be stowed on brackets inside a toolbox or curbside cabinet. The brackets shall be provided with retaining devices to secure the tools.

* 3.6.12.11 Hydraulic hose reel. When specified (see 6.2), a heavy duty, retractable hydraulic hose reel and hose shall be furnished. The hose reel shall be furnished with a mounting plate secured in the vehicle cabinet or in an accessible location on the vehicle platform. The hose reel shall be furnished with a weatherproof cover for protection when not in use. The hose reel shall be capable of withstanding not less than 110 percent of the hydraulic system working pressure. The hose is intended for use in the aerial basket and thus the hose and hose reel shall be capable of meeting the dielectric properties specified in 3.6.3. The hose reel shall be capable of retraction. The hose reel shall be marked stating that the hose shall be walked back to the reel to prevent damage. Hose fittings shall be of the quick-disconnect, dripless type and shall be compatible with the tool outlets specified in 3.6.12.1 and 3.6.12.1.1 and with the tools specified in 3.6.12.3 through 3.6.12.9. The following accessories shall be furnished:

(a) Flow control valve(s)
(b) Line check valve(s)
(c) Hose ball stop
(d) Four-way roller hose guide
(e) Dual hose, 15 m (50 feet) in length, with dust caps.

3.6.13 Safety. All equipment or exposed portions of the equipment that create extreme temperatures and that can be damaged by inclement weather, and all rotating or reciprocating parts which are of such nature or so located as to become a hazard to operating personnel, shall be insulated, fully enclosed, or properly guarded.

* 3.6.14 Stress analysis. A stress analysis for each class and size of vehicle to be furnished under the contract shall be performed. The stress analysis shall include the weight of all major items permanently installed on

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or within the vehicle, and shall show the location of the center of gravity of each component relative to the vehicle axles (travel and critical working positions). In addition, the analysis shall include the value of stresses at the base of the boom; area between the center of gravity of each component relative to the vehicle axles (travel and critical working positions). In addition, the data shall include the value of stresses at the base of the boom; area between the extension boom section and the basket; and the center of gravity of each section of the boom. The analysis shall verify the factors of safety specified herein, and shall verify that the yield point of the material used is not exceeded with the specified load on the basket.

3.7 Dissimilar metals. All dissimilar metals throughout the vehicle shall be insulated from one another to prevent galvanic or electrolytic action.

3.8 Lubrication system. A lubrication system shall be furnished with fittings for all bearings and other working parts which require relubrication for continued service. Vehicles shall be lubricated, prior to delivery, with applicable grade of lubricant for the ambient air temperature encountered at the delivery point.

3.9 Workmanship. The quality and workmanship of all component parts, including fabricated parts, machined parts and castings shall be commensurate with sound manufacturing standards for similar equipment. All metal and plastic materials used shall be free from delamination defects, and shall be prime material. Improper manufacturing or assembling practices and defective components or parts and assemblies which have been modified to overcome deficiencies shall not be furnished.

* 3.9.1 Welding. Adequate preparation of metal prior to welding of the body aerial tower components shall be provided. Surface of parts to be welded shall be free of rust, scale, paint, grease and other foreign matter. Welding rods (electrodes) shall be compatible with the metal being welded. All critical arc welds shall be magnetic particle or dye penetrant inspected. The following defects shall not be acceptable:

(a) Slug or plug welding
(b) Undersize fillet welds, overlaps, inadequate fusion and penetration, undercutting of weldments and vertical down welding
(c) Unremoved slag and spatter
(d) Cracks
(e) Spot, tack or intermittent welds for strength.

Weld penetration shall provide transference of maximum design stress through the base metal juncture. Intermittent welding shall be permissible when in accordance with 3.9.2 and 3.9.3.

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3.9.2 Workmanship specimens. Workmanship specimens shall be prepared by the manufacturer for the body and aerial tower components to simulate all welding conditions. The specimens shall represent the minimum acceptable weld quality and cleaning procedure to be used in production. Upon submittal of these specimens by the contractor, the quality level and identification shall be subject to approval by the Government. The specimens shall be prepared using the welding procedures to be employed in production. Specimens prepared to represent multiple pass welds shall be made in such a manner as to leave at least 50 mm (two inches) of each layer of weld metal exposed. Approval specimens shall be retained at the contractor's plant.

3.9.3 Welding procedures. The welding procedure shall be recorded prior to production of any weldment, but no procedure qualification other than workmanship specimens shall be required.

3.9.4 Factors. The factors listed below which are applicable shall be included in the recorded welding procedures:

(a) Joint sketch
(b) Joint type and preparation
(c) Base metal type
(d) Filler metal, size and type
(e) Position of welding
(f) Arc-voltage, current range and polarity
(g) Manual or machine
(h) Travel speed, machine only
(i) Pre-heat and post-heat.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examination and tests) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility for ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling

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inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

4.3 First vehicle preliminary inspections.

4.3.1 Welding procedure and workmanship specimen evaluation. Prior to start of fabrication of the subframe, turntable, pedestal and boom assemblies, an inspection shall be conducted by the Government to evaluate workmanship specimens and welding procedures.

4.3.1.1 Macro-etching. Workmanship specimens shall be cross sectioned and macro-etched. Both parts of the specimens shall be made available to the Government representative for examination.

4.3.2 In-process inspection of body and components. The first complete assembled body and components (sub-frame, body, turntable, outriggers, boom assemblies and basket or aerial platform) shall receive an in-process inspection by the Government prior to prime painting or application of protective coatings. The inspection shall be conducted to evaluate workmanship and to determine conformance of structural members to contractual requirements.

* 4.3.2.1 Structural inspection. The vehicle shall be tested in accordance with 4.4.7 and 4.4.8 prior to painting. After the tests, dye penetrant or magnetic particle inspection shall be performed on the critical areas of welds again prior to painting. A water-soluble dye penetrant inspection shall be performed on all critical plastic areas. The water-soluble dye penetrant or magnetic particle inspections shall provide means to detect crack indications after dynamic and static test loads.

4.4 First production vehicle inspection. The first production vehicle of each class and size produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformance to the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

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4.4.1 Weight and payload capacity. The first production vehicle shall be weighed to determine curb weight and distribution of curb weight on front and rear axles. The imposed loading on the front and rear axles will be computed to ascertain that the vehicle is adequate for the intended loading. Distribution of weight will be computed by using the curb weight, the operator weight at 80 kg (175 pounds), and a payload considered to be evenly distributed throughout the length of the vehicle body, to provide the specified GVW.

* 4.4.2 Road test. The vehicle, for 4x2 and 6x4 models and for models built to all wheel drive while on the chassis manufacturer's production line shall be road tested by the contractor with payload. The road test shall be for not less than 16 km (10 miles) at speeds up to 84 km/h (52 mph). The vehicle, for all wheel drive models qualifying under 3.1.7.14 shall be road tested with and without payload. Payload shall be distributed first, so that the front axle is loaded to GAWR and the vehicle is loaded to GVW and second, with the rear axle loaded to GAWR and the vehicle loaded to GVW. The road test for each of the three conditions shall be for not less than 48 km (30 miles) at speeds up to 84 km/h (52 mph) over highways and gravel roads and for not less than 8 km (5 miles) at speeds up to 24 km/h (15 mph) over cross country terrain with ground and grade requiring all wheel drive. During the loaded and empty portions of the road test, the brakes shall be applied firmly, bringing the vehicle to a sudden stop not less than 5 times during each portion of the road test. During the road test, the front and rear suspension and the drivetrain shall be periodically inspected for interference and contact with other vehicle components. Abnormal contact of the drivetrain or suspension components with other components shall be cause for rejection. Front spring bumpers shall not make contact with frame stops except under the most extreme cross country conditions. Operational or mechanical failures of the vehicle components during the road test shall be cause for rejection. Failure includes permanent deformation as well as breakage.

4.4.3 Upsetting and range of movement test. With the truck resting on approximately level ground, without payload, and not fastened to any artificial base, the basket or aerial platform shall be loaded with twice the specified payload (see 3.6.5 and 3.6.5.2). The basket or aerial platform shall be extended to its full reach and exercised through the full range of traverse and elevations indicated in 3.3.3. The inner dual tire shall be observed for conformance to the stability requirements of 3.3.3 with twice the load (see 3.6.5). The test shall be repeated with one and one half times the specified payload (see 3.6.5) and the outriggers and all tires observed for conformance to the stability requirements of 3.3.3 with one and one half times the load.

4.4.3.1 Stability on slopes test. The tests specified in 4.4.3 shall be repeated, except the truck shall be resting on a 5 degree slope, not leveled, in positions specified in 3.3.3.1.

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* 4.4.4 Wind test. With the vehicle not loaded, with the outriggers extended and the rated capacity placed in the basket or aerial platform, the boom shall be extended to maximum height and shall be rotated to that position in which the greatest area of the basket or aerial platform is parallel to the longitudinal centerline of the vehicle. The specified wind load, generated by artificial means or by an equivalent simulated load, shall be applied to the vehicle from either side (see 3.3.4).

4.4.5 Speed test. With the vehicle resting on approximately level ground, the basket or aerial platform shall be loaded with specified load (see 3.6.5 and 3.6.5.2). The basket shall be traversed and elevated to determine conformance to the requirements in 3.3.5 and 3.3.6.

* 4.4.6 Creep test. With the vehicle longitudinal on a 5 percent grade, without leveling the vehicle, and with specified load (see 3.6.5 and 3.6.5.2) on the basket or aerial platform, the basket shall be extended to the side with the boom at 45 degrees and the hydraulic fluid shall be stabilized at or near ambient temperature. The position shall be held for at least 1 hour for maximum horizontal creep of 25 mm (1 inch) per hour and vertical settling of 50 mm (2 inches) per hour, to determine conformance to requirements of 3.3.7.

* 4.4.7 Dynamic conditions. With two times the specified payload (see 3.6.5 and 3.6.5.2) in or suspended from the basket or aerial platform floor, the boom shall be extended through various positions, as deemed necessary by the Government inspector; and with the turntable being rotated 360 degrees, for at least 1/2 hour, the soluble penetrant and magnetic particle inspection areas shall be inspected for any evidence of cracks. Any evidence of cracks or deformation shall be cause for rejection of the vehicle.

4.4.8 Static conditions. With the boom in its most vertical position, and with three times the specified load (see 3.6.5 and 3.6.5.2) placed in or suspended from the basket or aerial platform, soluble penetrant and magnetic particle inspection areas shall be inspected for any evidence of cracks. Any evidence of cracks or deformation shall be cause for rejection of the vehicle. Dye penetrant and magnaflux inspection shall be conducted prior to the first production vehicle inspection. The tests in 4.4.7 and 4.4.8 need not be repeated at the first production vehicle inspection provided records of the tests and dye penetrant and magnaflux inspection, witnessed by the Government, are presented at the time of the first production vehicle inspection.

4.4.8.1 Critical areas. At least the following critical areas shall be inspected by dye penetrant or magnetic particle:

(a) Outriggers, and attaching areas
(b) Turntable; base, trunnion and yoke at attachment of boom and struts
(c) Boom at cylinder clevis brackets
(d) Structural steel of boom
(e) Yoke at attachment to basket and side brackets.

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* 4.4.9 Dielectric tests. The vehicle shall be tested as specified in the Qualification Test of ANSI A92.2, for Category A and B aerial devices for a 69,000 volts (69 kV) unit rating.

4.4.10 Performance test. The vehicle shall be subjected to, but not limited to, the following tests:

(a) Check to determine whether boom assembly can be elevated without outriggers;
(b) Determine if outriggers have locked automatically when in an extended position and in a retracted position;
(c) Operate equipment using each set of controls separately;
(d) Check to ensure that the lower controls override the basket or aerial platform controls;
(e) Check to ensure that the entire system can be operated smoothly and without jerk motions in any position;
(f) Check to determine that the basket or aerial platform remains level in all positions with twice the specified load in the basket or aerial platform;
(g) Check to determine that the basket or aerial platform is safely lowered from maximum reach position with rated payload on the basket or aerial platform floor to ground level with the engine not operating;
(h) Check the remote control starter and cutoff on the basket or aerial platform to ensure starting and stopping the engine with the specified payload in the basket or aerial platform;
(i) Check overload capabilities by attempting to lift in excess of two times rated capacity of the basket or aerial platform (see 3.6.5 and 3.6.5.2) without increasing hydraulic pressure with boom horizontal and extended;
(j) Check the hydraulic system including the pump at 10 percent above the pump manufacturer's maximum operating pressure setting for normal workload conditions.
(k) Check hydraulic fluid temperature after system reaches steady state condition to determine conformance to 3.6.8.6.

Upon completion of all testing, the hydraulic system shall be adjusted to limit the capability at maximum horizontal reach to not more than 1.25 times the specified load capacity (see 3.6.5 and 3.6.5.2).

4.4.11 Body treatment and painting. The certification regarding body cleaning, treating, prime painting and salt spray resistance testing, as required by MIL-STD-1223, shall be presented to Government representatives for examination and approval.

4.4.12 Radio frequency suppression verification. The vehicle manufacturer shall indicate on the questionnaire (commercial vehicle engineering data) if the vehicle will be suppressed to limit electromagnetic radiation in accordance with SAE J551.

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4.4.13 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample. The production sample shall be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in an as new condition for the duration of the contract.

4.4.14 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action has been taken. In addition, subsequent vehicles that do not conform to the accepted production sample vehicle shall be rejected until corrected.

* 4.5 Availability of ANSI A92.2. The contractor shall make available to the Government at the point of final acceptance and test, current copies of ANSI A92.2. The document shall be surrendered to the Government for personal use of Government representatives in conjunction with the contract for the duration of the contract.

4.6 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting performance requirements contained herein. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required by 3.4.25 have been accomplished.

5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment, from manufacturer's plant to initial receiving activity, in accordance with the manufacturer's standard commercial practice.

6. NOTES

* 6.1 Intended use. The vehicles covered by this specification are intended for general nontactical use by the Government for electrical distribution lines not exceeding 46,000 volts with the basket liner installed, for electric utility line maintenance, replacement of electric lamps, tree trimming, painting, window washing, and many other types of overhead construction and maintenance work.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, size and model of vehicle required (see 1.2).
* (c) Issue of DoDISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

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total reserve capacity rating and the total cold cranking rating at -18 C

(d) Identification of appropriate service for painting (see 3.1.1.1).
(e) Identification of appropriate service for marking (see 3.1.1.2).
(f) Concealed marking, if required (see 3.1.1.2).
(g) Rustproofing, if required (see 3.1.1.3).
(h) Tropical rustproofing, if required (see 3.1.1.3).
(i) A trailer electrical cable, if required (see 3.1.1.7).
(j) Electric trailer brake controller, if required (see 3.1.1.8).
(k) A trailer air brake system, if required (see 3.1.1.9).
(l) Silicone brake fluid, if required for classes A, B and C (see 3.1.1.14).
(m) Satisfactory operation on JP-4, JP-5 and JP-8 if required (see 3.4.1.1).
(n) Gasoline engine, if required (see 3.4.1.2).
(o) Silicone rubber radiator and heater hoses, if required (see 3.4.1.8).
(p) Power plant heaters and fuel warmer, if required (see 3.4.1.9).
* (q) Vernier throttle control, if required (see 3.4.1.10).
* (r) Starting motor circuit breaker, if required (see 3.4.2.1).
(s) Alternator capacity, if other than as specified (see 3.4.23).
(t) Automatic transmission, if required (see 3.4.5.3).
(u) Two-speed axle, if required for 4x2 and 6x4 model vehicles (see 3.4.9.2).
(v) Traction control, if required (see 3.4.9.3).
(w) Wide base type front and rear tires and wheels, if required (see 3.4.10).
(x) Disc type wheels, if required (see 3.4.10).
(y) Bias ply tires, if required for class C, D, E and F vehicles (see 3.4.10.1).
* (z) Low profile tires, if required (see 3.4.10.1).
(aa) A carrier for a spare tire assembly, if required (see 3.4.10.3).
(ab) Spare wheel or rim, if required (see 3.4.10.4).
(ac) Spare tire assembly, if required (see 3.4.10.5).
(ad) Full air brakes for class A, B and C vehicles, if required (see 3.4.11.2).
(ae) Brake controls for use from a towing vehicle, if required (see 3.4.11.4).
* (af) Tilting hood and fender assembly, if required (see 3.4.12).
(ag) Individual driver and passenger seats, if required (see 3.4.12.1).
* (ah) Air ride driver's seat, for vehicles with air brakes, if required (see 3.4.12.1).
(ai) Crew cab, if required (see 3.4.12.2).
* (aj) Intermittent windshield wipers, if required (see 3.4.14).
(ak) Tools, if required (see 3.4.16.1).
* (al) Remote control curbside rearview mirrors, if required (see 3.4.20).
* (am) Heated outside rearview mirrors, if required (see 3.4.20).
(an) Air-operated horn, in addition, if required (see 3.4.21).
(ao) A front mounted winch, if required (see 3.4.23).
(ap) Winch reel, if required (see 3.4.23.4).

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(aq) Air conditioning, if required (see 3.4.24).
* (ar) Cooling system protection down to -54°C (-65°F), if required (see 3.4.25.1).
(as) Water cask, if required (see 3.5.2.8).
(at) Uninsulated guardrail type platform, if required for size 60 vehicles (see 3.6.5.2).
(au) Hydraulic fluid conforming to MIL-H-83282, if required (see 3.6.8.4.1).
* (av) Rechargeable electrical system, if required in addition to PTO (see 3.6.9).
* (aw) Auxiliary engine, if required in addition to PTO (see 3.6.10).
(ax) An engine hour meter for auxiliary engine, if required (see 3.6.10.1(g)).
(ay) A separate auxiliary engine fuel tank, if required (see 3.6.10.6.1).
(az) An air compressor, if required (see 3.6.11).
(ba) If a hydraulic ground tool outlet is not required (see 3.6.12.1).
(bb) If a hydraulic basket tool outlet and hose are not required (see 3.6.12.1.1).
(bc) A hydraulic capstan hoist, if required (see 3.6.12.3).
(bd) A hydraulic impact wrench and drill, if required (see 3.6.12.4).
(be) A hydraulic chain saw, if required (see 3.6.12.5).
(bf) A hydraulic chain saw, pole mounted, if required (see 3.6.12.6).
(bg) A hydraulic circular saw, pole mounted, if required (see 3.6.12.7).
(bh) A hydraulic pruner, pole mounted, if required (see 3.6.12.8).
(bi) A hydraulic hose reel, if required (see 3.6.12.11).

* 6.3 Performance prediction. SAE Truck Ability Prediction Procedure computations and computations for low speed and maximum geared speed will be required by the contract. The SAE Work Sheet Item 1 should include vehicle model number, engine model number, and vehicle type, class and model. Unless other conditions are cited in the contract, computations should be made for normal atmospheric pressure, normal ambient air temperature, and still dry air. The factors to be used in predicting truck ability (see 3.3.1.1) are established as follows for the corresponding SAE Truck Ability Prediction Procedure Tables:

Table 1	- Tire Factor. This factor must relate to the size of tires furnished by the contractor in accordance with this specification.
Table 2	- Altitude Factor. 1.00
Table 3	- Rolling Factor. 1.613
Table 4	- Area Factor. 0.173
Table 5	- Velocity Factor. 250.0
Table 6	- Altitude Factor. 1.00

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Table 7	- Chassis Friction Horsepower. Use applicable power unit GVW (to nearest, higher, 454 kg (1,000 pounds)) and the engine rpm (to nearest 100 revolutions) which is required for 80 km/h (50 mph) geared speed. For GVW and engine speed beyond the range of this table, factors shall be extrapolated.
Table 8	- Grade Factor. 0.75
Table 8A	- Correction Factor. Not required.
Table 9	- Road Factor. 0.0.

* 6.4 Subject term (key word) listing.

Aerial platform
Basket leveling system
Fiberglass basket
Hydraulic system
Outriggers
Rechargeable electrical system
Rotating Turntable
Stirrup step

* 6.5 Identification of changes. The margins of this document are marked with asterisks (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, regardless of the marginal notations and relationship to the previous issue.

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